Understanding Multilateral Tariffs and Exchange Rates*

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Abstract

The standard prediction that an import tariff appreciates the tariffing country's currency comes from a two-country setting, where the bilateral exchange rate, effective exchange rate, and the terms of trade coincide. In a multilateral setting, these measures separate and respond differently to tariffs. Under symmetric trade relationships, we show that a unilateral tariff appreciates the effective exchange rate but can depreciate the home currency against untariffed bystanders when substitution across foreign suppliers is sufficiently elastic. Tariff retaliation can instead lead both counterparts to depreciate in effective terms. These results are robust to moderate asymmetry, but under strong asymmetry, tariffs on small trading partners can worsen the tariffing country's terms of trade and reduce welfare when foreign suppliers are sufficiently close substitutes. We use the framework to contrast exchange-rate responses during the 2025 tariff episode.

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1 Introduction

What happens to exchange rates when a country imposes a tariff? In the standard two-country model, a tariff shifts expenditure away from imports and improves the trade balance at the prevailing exchange rate. Under the Marshall–Lerner condition, external balance is restored through an appreciation of the tariffing country's currency (Dornbusch, 1974; Krugman, 1982). Because there is only one trading partner, the bilateral exchange rate is also the effective exchange rate. In the long run, in real terms, this same relative price also governs the terms of trade. Thus, in the simplest bilateral setting, the bilateral exchange rate, the effective exchange rate, and the long-run terms of trade all necessarily coincide.

These concepts separate once trade is multilateral. A tariff on one partner changes not only the relative price of foreign and domestic goods, but also relative prices across foreign suppliers. It can therefore affect the exchange rate against the target, the rates against untariffed bystanders, and the effective exchange rate differently. In any differentiable multi-country trade environment, we show that the first-order response can be represented as the adjustment of exchange rates to a vector of tariff-induced trade-balance disturbances. We define a multilateral Marshall–Lerner condition, implied by gross substitutability of trade balances in currency values, that ensures well-behaved adjustment. Crucially, we show that even under this condition, the sign of an individual bilateral exchange-rate response is not determined. A tariff therefore need not appreciate the tariffing country's currency against every partner.

We first answer this question in a general trade environment under symmetry among trading partners. We show that any vector of multilateral tariff changes decomposes into average protection and relative treatment. Average protection uniquely determines the change in the nominal effective exchange rate, while relative treatment determines the relative changes in bilateral exchange rates. Under a unilateral tariff, these two components reinforce each other against the targeted partner but work in opposite directions against untariffed bystanders. The response against bystanders is therefore ambiguous and depends on the structure of the trade environment.

To characterize these counteracting effects, we develop a nested CES Armington model that distinguishes substitution between domestic and foreign goods from substitution across foreign suppliers. Substitution toward domestic goods supports the conventional appreciation, while substitution across foreign suppliers shifts expenditure toward untariffed countries and strengthens their currencies relative to the tariffing country. Allowing these two margins to differ, we show that the tariffing country can depreciate against an untariffed bystander when cross-foreign substitution is sufficiently strong relative to home-foreign substitution,

but, under weak home bias, this outcome cannot arise with a single elasticity across domestic and foreign varieties. Empirical estimates distinguishing cross-foreign from home-foreign substitution support treating these margins separately (Broda and Weinstein, 2006; Feenstra et al., 2018; Li et al., 2022).

The same mechanisms extend naturally from unilateral tariffs to retaliation and multilateral trade wars. When the target retaliates with an equal tariff, the average-protection effects on the bilateral exchange rate cancel, while the relative-treatment effects reinforce each other against third countries. As a result, the threshold for depreciation against bystanders is substantially lower than under a unilateral tariff. In a multilateral trade war, in which every partner retaliates against the tariffing country, the same condition leads it to depreciate against every partner.

Our trade framework is deliberately stylized and is not intended to provide a quantitative account of the 2025 tariff episode, but it nonetheless offers a new perspective on the exchange-rate movements during this period. The dollar's depreciation following the tariff announcements was initially interpreted in part as evidence of higher risk premia, declining convenience yields, and concerns about the dollar's global role (Itskhoki and Mukhin, 2025; Kalemli-Özcan et al., 2026). Applying our framework to the broad features of the episode, we show that dollar depreciation need not originate outside the trade channel, particularly when retaliation is expected. Consistent with this explanation, Ostry et al. (2025) document that the dollar tends to appreciate following unilateral US tariffs, depreciate when trading partners retaliate, and remain comparatively stable against the renminbi during US–China tit-for-tat episodes. They hypothesize that dominant-currency pricing may help explain this pattern. Our results show that the same pattern can arise from canonical trade substitution and diversion effects once trade and exchange-rate adjustment is considered in a multilateral setting.

In a general asymmetric setting, where economies differ in size, trade shares, and productivities, exchange-rate responses no longer admit simple analytical expressions that characterize when one force dominates another. We show that the main qualitative results are robust to moderate asymmetry, although under strong asymmetry some aggregate conclusions can reverse. In particular, when cross-foreign substitution is very strong and the targeted country accounts for only a small share of the tariffing country's trade, diversion toward more important bystanders can dominate the average-protection effect and worsen the aggregate terms of trade. We examine these reversals numerically in the three-country model and find that they require relatively extreme configurations of country size and trade shares. This possibility is therefore less relevant for understanding the 2025 tariff episode, when the United States imposed substantial protection on its major trading partners.

These reversals apply to discriminatory tariffs, not to uniform protection. Under general conditions allowing country sizes, trade shares, home bias, and preference parameters to differ, we prove that a small uniform tariff imposed from a balanced free-trade baseline appreciates the tariffing country's currency weakly against every partner under the multilateral Marshall–Lerner condition. It therefore improves the aggregate terms of trade and raises welfare to first order. This result recovers the classical terms-of-trade argument for uniform protection while showing why its partner-specific counterpart can fail. Uniform protection eliminates relative treatment across suppliers and switches off the cross-origin diversion mechanism that drives the reversals.

Our paper builds on the classical analysis of tariffs under flexible exchange rates, which identifies the conditions under which tariffs lead to exchange-rate appreciation (Lerner, 1936; Metzler, 1949; Mundell, 1961; Dornbusch, 1974; Eichengreen, 1981; Krugman, 1982). A large open-economy macroeconomics literature extends this analysis to environments with nominal rigidities, monetary policy, currency invoicing, and alternative uses of tariff revenue (Lindé and Pescatori, 2019; Barattieri et al., 2021; Erceg et al., 2023; Bergin and Corsetti, 2023; Werning et al., 2025). We instead revisit the flexible-price benchmark in a multilateral setting, where tariffs affect not only the importing country's aggregate exchange rate but also relative exchange rates across trading partners. Our bystander results are related to the literature on third-country effects of preferential trade policy, particularly the roles of trade diversion and competition among exporters (Viner, 1950; Mundell, 1964; Bond and Syropoulos, 1996; Bagwell and Staiger, 1999, 2002). We show how these forces affect bilateral and effective exchange rates when the importing country also produces a competing good. Finally, our multilateral Marshall–Lerner condition draws on the classical literature on stability in multiple markets (Hicks, 1939; Mosak, 1944; Metzler, 1945; Arrow and Hurwicz, 1958), applying these conditions to the exchange-rate response to tariffs in a multilateral setting.

Our results also relate to the empirical literature on the exchange-rate effects of tariffs, including the 2025 episode (Itskhoki and Mukhin, 2025; Kalemli-Özcan et al., 2026; Ostry et al., 2025) and earlier episodes (Furceri et al., 2019; Matveev and Ruge-Murcia, 2021; Jeanne and Son, 2024). Our analysis also offers a complementary perspective on the empirical literature estimating tariff pass-through into import prices (Amiti et al., 2019; Fajgelbaum et al., 2020; Cavallo et al., 2021). This literature typically uses dollar-denominated import prices as a proxy for tariff incidence. In a multilateral setting, however, exchange-rate movements provide an additional margin of incidence through changes in relative prices. An exporter whose dollar price is unchanged may still experience a decline in the home-currency value of its receipts when the dollar depreciates. Assessing tariff incidence from import prices

alone can therefore overlook adjustment through exchange rates, a margin that may have been particularly relevant during the broad dollar depreciation in 2025.

This paper is organized as follows. Section 2 develops the general multilateral framework, derives the multilateral Marshall–Lerner condition, and decomposes tariff changes into average protection and relative treatment. Section 3 gives this decomposition economic content in a nested-CES model and derives the main results for unilateral tariffs. Section 4 extends the analysis to bilateral retaliation, multilateral trade wars, and preferential tariffs. Section 5 considers asymmetric trade relationships, examines how the results change away from symmetry, and applies the framework to the 2025 tariff episode. Section 6 concludes, with derivations, proofs, and additional robustness results provided in the appendices.

2 Tariffs and Exchange-Rate Adjustment

2.1 The bilateral benchmark

Consider two countries, A and B. Let e_{AB} denote units of A's currency per unit of B's, so that a rise is a depreciation of A, and write $\nu \equiv \log e_{AB}$. Country A levies an ad valorem tariff τ on imports from B.

Each country's trade balance TB_i is the value of its exports minus the value of its imports, both at border (pre-tariff) prices and both expressed in one fixed common unit of account. Tariff revenue is a domestic transfer: it is counted neither as export revenue nor as an import payment, although its rebate or other use may affect income, quantities and therefore the trade balance. In the nested-CES model of Section 3 it is rebated lump-sum. Let v_l denote the common-unit value of one unit of currency l. We select A as numeraire and normalize $v_A = 1$, so these trade-balance values are numerically equal to values expressed in A's currency. This convention is used throughout this section and in the proof of Lemma 2.3, and all derivatives in this section are evaluated at the balanced baseline unless otherwise indicated.

Equilibrium with balanced trade is $\mathrm{TB}_A(\nu, \tau) = 0$. Linearizing,

$$d\,\mathrm{TB}_A = \mathrm{TB}_{A\nu}\,d\nu + \mathrm{TB}_{A\tau}\,d\tau = 0 \quad \Longrightarrow \quad d\nu^* = -\frac{\mathrm{TB}_{A\tau}}{\mathrm{TB}_{A\nu}}\,d\tau, \tag{2.1}$$

where $\mathrm{TB}_{A\nu} \equiv \partial \mathrm{TB}_A/\partial\nu$ holds the tariff fixed and $\mathrm{TB}_{A\tau} \equiv \partial \mathrm{TB}_A/\partial\tau$ holds ν fixed. At fixed ν, the tariff produces a positive trade-balance disturbance, $\mathrm{TB}_{A\tau} > 0$, when the tariff-induced reduction in A's import payments exceeds any tariff-induced reduction in its export receipts, with tariff revenue treated according to the convention above. Conditional on $\mathrm{TB}_{A\tau} > 0$,

Eq. (2.1) implies an appreciation if and only if $\mathrm{TB}_{A\nu} > 0$, that is, if and only if a depreciation improves the trade balance.

To express $\mathrm{TB}_{A\nu}$ in elasticities, let Q_X and Q_M be A's export and import quantities, and hold border prices fixed in the producer's currency, as in the nested-CES model, with units chosen so that each border price is one. A's export receipts are then Q_X and its import payments $e_{AB}Q_M$, both in A's currency, so $\mathrm{TB}_A = Q_X - e^{\nu}Q_M$. Define

$$\varepsilon_X \equiv \frac{\partial \log Q_X}{\partial \nu}, \qquad \varepsilon_M \equiv -\frac{\partial \log Q_M}{\partial \nu},$$

at fixed τ and at the balanced baseline. Both are positive when a depreciation raises export volume and lowers import volume. Let $X \equiv Q_X = e_{AB}Q_M$ be the common baseline border value of exports and imports. Differentiating, $\mathrm{TB}_{A\nu} = Q_X\varepsilon_X - e_{AB}Q_M(1-\varepsilon_M) = X(\varepsilon_X + \varepsilon_M - 1)$: the -1 is the valuation effect of a depreciation on the payment for each imported unit. The condition $\mathrm{TB}_{A\nu} > 0$ is therefore the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. Figure 1 illustrates; the parallel shift it shows is a first-order approximation around the baseline.

Remark 2.1 (Robustness of the two-country result). The sign $d\nu^*/d\tau < 0$ requires two conditions:

(i) at fixed ν, the tariff-induced fall in import payments is larger than any tariff-induced fall in export receipts, so $\mathrm{TB}_{A\tau} > 0$;

(ii) the Marshall–Lerner slope is positive, $\mathrm{TB}_{A\nu} > 0$.

Neither condition holds in every model or under every treatment of tariff revenue. Appendix D.1 states sufficient conditions for each, under which $d\nu^*/d\tau < 0$. In the two-country model, a tariff therefore appreciates the tariffing country's currency whenever both conditions hold. The rest of the paper shows that with three or more countries this is no longer enough.

2.2 Multilateral adjustment

Now consider N countries, indexed by $\mathcal{I} = \{1, \ldots, N\}$, and let $\mathcal{I}_{-A} = \mathcal{I} \setminus \{A\}$. Take A's currency as numeraire. The $N-1$ bilateral rates e_{Aj}, $j \in \mathcal{I}_{-A}$, span all exchange-rate changes. Trade balances sum to zero (Walras' law), which leaves $N-1$ independent trade-balance conditions, indexed by $i \in \mathcal{I}_{-A}$:

$$\mathrm{TB}_i\Big(\{e_{Aj}\}_{j\in\mathcal{I}_{-A}}, \{\tau_{jk}\}_{(j,k)\in\mathcal{T}}\Big) = 0, \qquad i \in \mathcal{I}_{-A},$$

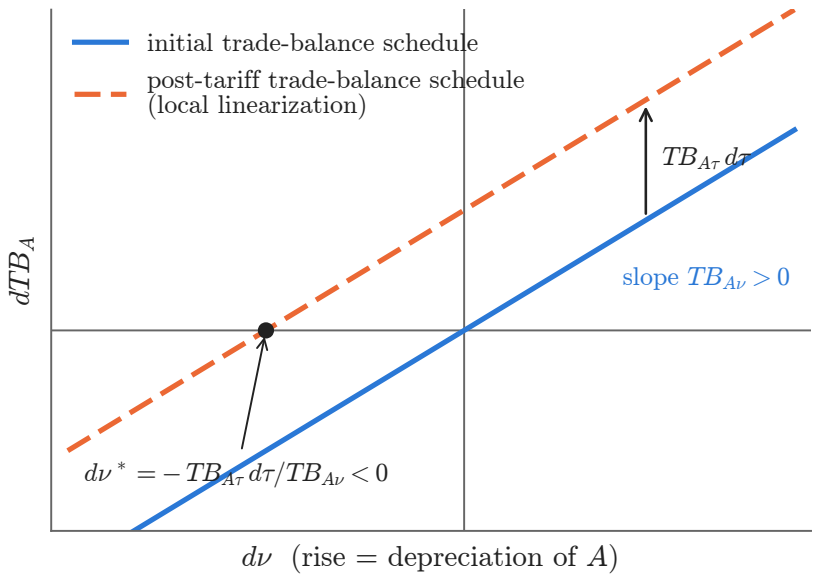


Figure 1: Bilateral tariff and exchange-rate adjustment. At a fixed exchange rate, the tariff shifts the trade-balance schedule upward. When the Marshall–Lerner slope is positive, equilibrium is restored through an appreciation. Schematic.

where τ_{jk} is country j's ad valorem tariff on imports from k and $\mathcal{T} \equiv \{(j,k) \in \mathcal{I} \times \mathcal{I} : j \neq k\}$ is the set of all $N(N-1)$ ordered country pairs. Stack the bilateral rates as $\nu = (\nu_l)_{l \in \mathcal{I}_{-A}}$ with $\nu_l = \log e_{Al}$, in increasing order of l, and the tariffs as $\tau = (\tau_{jk})_{(j,k)\in\mathcal{T}}$, in lexicographic order of (j,k). Linearizing,

$$J\, d\nu + G\, d\tau = 0, \qquad J_{il} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \nu_l}\right|_{\tau}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{jk}}\right|_{\nu}, \tag{2.2}$$

for $i, l \in \mathcal{I}_{-A}$ and $(j,k) \in \mathcal{T}$, so $J \in \mathbb{R}^{(N-1)\times(N-1)}$ and $G \in \mathbb{R}^{(N-1)\times N(N-1)}$. Consider a tariff change along a fixed direction,

$$d\tau = a\, ds, \qquad a \in \mathbb{R}^{N(N-1)}, \quad \|a\|_2 = 1,$$

where the unit norm serves only to normalize the scalar s. Define $F \equiv Ga$, the marginal vector of trade-balance disturbances per unit increase in s, evaluated at unchanged exchange rates. Substituting into Eq. (2.2),

$$J \frac{d\nu}{ds} + Ga = 0.$$

Provided J is nonsingular, therefore,

$$\boxed{\frac{d\nu}{ds} = (-J)^{-1}F.} \tag{2.3}$$

Equation (2.3) separates the trade-balance effects of the tariff at unchanged exchange rates, F, which include expenditure switching, tariff revenue and income effects, from the exchange-rate movements required to restore external balance, $(-J)^{-1}$. In the bilateral case both are scalars, and Eq. (2.3) reduces to Eq. (2.1). With $N \geq 3$, however, the elements of F need not have the same sign: a tariff imposed by A on B can improve a bystander C's trade balance at unchanged exchange rates. The response of any bilateral exchange rate, including ν_C, therefore depends on the full vector of tariff-induced trade-balance disturbances, not on its own element alone.

2.3 A multilateral Marshall–Lerner condition

Definition 2.2 (Multilateral Marshall–Lerner condition, MML)**.** Let $i, l \in \mathcal{I}_{-A}$. The adjustment mechanism satisfies MML if $J_{il} \geq 0$ for every $l \neq i$, and $(-J)^{-1}$ exists with $(-J)^{-1} \geq 0$ componentwise, that is, $[(-J)^{-1}]_{il} \geq 0$ for all i, l. Equivalently, $-J$ is a nonsingular M-matrix.

Under MML, writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_{j \in \mathcal{I}_{-A}} \vartheta_{ij} F_j, \qquad i \in \mathcal{I}_{-A}. \tag{2.4}$$

Every disturbance enters every bilateral response with a nonnegative weight. Hence, componentwise, if $F \geq 0$ then $d\nu/ds \geq 0$, and if $F \leq 0$ then $d\nu/ds \leq 0$, where a positive $d\nu_i/ds$ is a depreciation of A against i. This is the multilateral counterpart of the two-country result.

A primitive sufficient condition for MML is gross substitutability of trade balances, which is stated in terms of the full $N \times N$ Jacobian with respect to currency values. Let $\chi_l \equiv \log v_l$, $l \in \mathcal{I}$, be the log common-unit value of currency l. Since $e_{Aj} = v_j / v_A$,

$$\nu_j = \chi_j - \chi_A, \qquad j \in \mathcal{I}_{-A}.$$

Define $\widetilde{J}_{il} \equiv \partial \mathrm{TB}_i / \partial \chi_l$ for $i, l \in \mathcal{I}$, at fixed tariffs. If each trade balance is homogeneous of degree one in currency values, Euler's identity gives $\sum_{l \in \mathcal{I}} \widetilde{J}_{il} = \mathrm{TB}_i = 0$ at the balanced baseline, and Walras' law, $\sum_{i \in \mathcal{I}} \mathrm{TB}_i \equiv 0$, gives $\sum_{i \in \mathcal{I}} \widetilde{J}_{il} = 0$ for every $l \in \mathcal{I}$. Because χ_A is fixed by the normalization $v_A = 1$, a change in ν_l is a change in χ_l alone, so the reduced

Jacobian of Eq. (2.2) is $J = (\widetilde{J}_{il})_{i,l\in\mathcal{I}_{-A}}$. Trade balances are gross substitutes if

$$\widetilde{J}_{il} \geq 0 \qquad \text{for all } i, l \in \mathcal{I},\ l \neq i,$$

that is, if an appreciation of currency l, a rise in χ_l, weakly improves every other country's trade balance.

Lemma 2.3. *If each trade balance is homogeneous of degree one in currency values, $\widetilde{J}_{il} \geq 0$ for all $i, l \in \mathcal{I}$ with $l \neq i$, the reduced matrix $J = (\widetilde{J}_{il})_{i,l\in\mathcal{I}_{-A}}$ is irreducible, and $\widetilde{J}_{iA} > 0$ for at least one $i \in \mathcal{I}_{-A}$, then MML holds.*

Lemma 2.3 is the trade-balance analogue of the gross-substitutes stability results of Hicks (1939), Mosak (1944), Metzler (1945) and Arrow et al. (1959). MML ensures that multilateral adjustment is well behaved, and a disturbance vector of one sign produces exchange-rate responses of the same sign. When F contains both positive and negative elements, however, MML alone does not determine the sign of an individual bilateral exchange-rate response. Positive own-exchange-rate trade-balance slopes are therefore not sufficient to sign bilateral responses in a multilateral setting.

2.4 Symmetric trading partners

Equation (2.3) holds at any balanced baseline. When the tariffing country's partners are symmetric, it can be solved using two eigenspace-specific eigenvalues of $-J$ and two of $G^{(A)}$.

Restrict attention to tariff changes imposed by A, $t = (t_j)_{j\in\mathcal{I}_{-A}}$ with $t_j \equiv d\tau_{Aj}$, and let $G^{(A)} \in \mathbb{R}^{(N-1)\times(N-1)}$ be the corresponding restriction of the tariff-incidence matrix,

$$G^{(A)}_{ij} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{Aj}}\right|_{\nu}, \qquad i, j \in \mathcal{I}_{-A},$$

so that $J\,d\nu + G^{(A)}t = 0$. Assume that the primitives and the balanced baseline allocation are invariant to every relabeling of A's $N-1$ partners. This implies that, for every $(N-1)\times(N-1)$ partner-permutation matrix P,

$$PJP' = J \qquad \text{and} \qquad PG^{(A)}P' = G^{(A)}.$$

A free-trade or common-external-tariff baseline is an example only when the partner primitives are also symmetric. We then call A's partners symmetric. With $\mathbf{1}$ the $(N-1)$-vector of ones, decompose

$$t = \bar{t}\,\mathbf{1} + \widetilde{t}, \qquad \bar{t} = \tfrac{1}{N-1}\mathbf{1}'t, \qquad \mathbf{1}'\widetilde{t} = 0. \tag{2.5}$$

We call $\bar{t}$ *average protection* and $\widetilde{t}$ *relative treatment*. Any permutation-invariant matrix has the form $aI + b\mathbf{1}\mathbf{1}'$, so the one-dimensional common subspace $\text{span}\{\mathbf{1}\}$ and the $(N-2)$-dimensional deviation subspace $\{v : \mathbf{1}'v = 0\}$ are eigenspaces of both $G^{(A)}$ and $-J$:

$$G^{(A)}\mathbf{1} = g_S\mathbf{1}, \quad G^{(A)}v = g_D v; \qquad (-J)\mathbf{1} = j_S\mathbf{1}, \quad (-J)v = j_D v \qquad (\mathbf{1}'v = 0).$$

The eigenvalues j_D and g_D each have multiplicity $N-2$.

We measure A's effective exchange rate by its nominal effective exchange rate (NEER) with baseline bilateral trade weights. With f_{ij} the border value of country i's imports from j,

$$w_{Aj} \equiv \frac{f_{Aj} + f_{jA}}{\sum_{k\neq A}(f_{Ak} + f_{kA})}, \qquad \sum_{j\neq A} w_{Aj} = 1,$$
$$E_A \equiv \prod_{j\neq A} e_{Aj}^{w_{Aj}}, \qquad d\nu_A^E \equiv d\log E_A = \sum_{j\neq A} w_{Aj}\, d\nu_j,$$

holding the baseline weights fixed for the local comparative static. This is the effective exchange rate throughout the paper. Under partner symmetry the bilateral trade weights are equal, $w_{Aj} = 1/(N-1)$, so in this section the NEER reduces to the simple average of the bilateral log exchange rates, $d\nu_A^E = \frac{1}{N-1}\sum_{j\neq A} d\nu_j$. For partners $j, k \in \mathcal{I}_{-A}$ the cross rate is $e_{jk} = e_{Ak}/e_{Aj}$, units of j's currency per unit of k's.

Proposition 2.4 (Average protection and relative treatment)**.** *Let partners be symmetric, $N \geq 3$, and $j_S, j_D \neq 0$, so that J is nonsingular. For any tariff vector t,*

$$d\nu = \lambda_{\text{avg}}\,\bar{t}\,\mathbf{1} + \lambda_{\text{rel}}\,\widetilde{t}, \qquad \lambda_{\text{avg}} = \frac{g_S}{j_S}, \quad \lambda_{\text{rel}} = \frac{g_D}{j_D}. \tag{2.6}$$

Hence

$$d\nu_A^E = \lambda_{\text{avg}}\bar{t}, \qquad d\nu_j - d\nu_A^E = \lambda_{\text{rel}}\widetilde{t}_j, \qquad d\log e_{jk} = d\nu_k - d\nu_j = \lambda_{\text{rel}}(t_k - t_j). \tag{2.7}$$

Moreover, under the permutation-symmetry assumptions above, MML is equivalent to $j_S > 0$ and $j_D \geq j_S$.

Average protection shifts all of A's bilateral log exchange rates by the same amount, so under symmetry it determines the NEER. Relative treatment has zero mean, leaves the NEER unchanged, and moves bilateral rates around the common component; it also determines the movements in cross rates among partners. The two components lie in complementary subspaces, $\mathbb{R}^{N-1} = \text{span}\{\mathbf{1}\} \oplus \{v : \mathbf{1}'v = 0\}$, and each is scaled by its own coefficient. With $N = 3$ these subspaces span a plane, which Fig. 2 uses to illustrate the decomposition.

Proposition 2.4 thus reduces the exchange-rate response to any tariff vector to two coefficients: λ_{avg}, through which average protection moves the common component of A's bilateral rates and hence its effective exchange rate, and λ_{rel}, through which relative treatment moves individual bilateral rates around that component and the cross rates among partners. The decomposition follows from partner symmetry alone and does not rely on the nested-CES structure or any other particular demand system. What remains undetermined are the signs and magnitudes of λ_{avg} and λ_{rel}, and hence the relative strength of the two components. Section 3 introduces a nested-CES model in which these coefficients depend on substitution between domestic and foreign goods and on substitution across foreign suppliers.

3 The Nested-CES Model

This section specializes the decomposition of Section 2.4 into average protection and relative treatment to a nested-CES trade model. In the model, η governs substitution between the domestic tradable and the import composite, and ρ governs substitution across foreign origins. We work locally, to first order around a symmetric, balanced, free-trade benchmark. The central result is that a unilateral tariff always appreciates the tariffing country's NEER and its currency against the target, but depreciates it against untariffed bystanders only when $\rho > N\rho_D$, where ρ_D is a domestic diversion margin defined below. With weak home bias, $\alpha_D \geq 1/N$, a specification with a single elasticity, $\rho = \eta$, cannot produce this reversal.

3.1 Production, preferences and equilibrium

There are $N \geq 2$ countries, $i \in \{1, \ldots, N\}$. Country A is the tariffing country, B is the targeted partner, and lowercase c denotes an individual untariffed bystander; uppercase C is reserved for the composite bystander of Section 3.5. Each country is endowed with L_i units of labor, mobile across sectors, and produces a country-specific tradable and a non-tradable with labor only, $Y_{Ti} = A_{Ti}L_{Ti}$ and $Y_{Ni} = A_{Ni}L_{Ni}$, with $L_{Ti} + L_{Ni} = L_i$. Normalizing each tradable's producer price to one in its own currency gives $w_i = A_{Ti}$ and $P_{Ni} = A_{Ti}/A_{Ni}$. Relative international producer prices therefore move only through exchange rates, and the bilateral exchange rate equals the bilateral terms of trade and the relative wage in a common currency.

Preferences are

$$C_i = C_{Ti}^{\alpha_T} C_{Ni}^{1-\alpha_T}, \quad C_{Ti} = \left[\alpha_D^{1/\eta} C_{Tii}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \quad M_i = \left[\textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho} C_{Tji}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}, \tag{3.1}$$

with $0 < \alpha_D < 1$, $0 < \alpha_T \leq 1$, $\eta, \rho > 0$ (read as Cobb–Douglas at unit elasticity), $\beta_{ij} > 0$ and $\sum_{j \neq i} \beta_{ij} = 1$. Country i levies an ad valorem tariff τ_{ij} on imports from j, so the consumer price is $p_{ij} = (1 + \tau_{ij}) e_{ij}$, and the import and tradable price indices are $P_{Mi} = [\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho}]^{1/(1-\rho)}$ and $P_{Ti} = [\alpha_D + (1 - \alpha_D) P_{Mi}^{1-\eta}]^{1/(1-\eta)}$. Income is

$$I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1+\tau_{ij}} s_{ij}}, \tag{3.2}$$

where s_{ij} is the tariff-inclusive expenditure share of income on j's good. Tariff revenue is rebated lump-sum and is a domestic transfer, so the external-balance conditions use flows at border prices:

$$\mathrm{TB}_i = \sum_{k \neq i} f_{ki} - \sum_{j \neq i} f_{ij}, \qquad f_{ij} = \frac{s_{ij} \tilde{I}_i}{1 + \tau_{ij}}, \tag{3.3}$$

with $\tilde{I}_i = e_{Ai} I_i$ income in A's currency. Under the normalization $v_A = 1$ of Section 2.1, values in A's currency equal values in the common unit of account, so Eq. (3.3) is the trade-balance measure of Section 2. With A's currency as numeraire, the $N - 1$ independent conditions $\mathrm{TB}_i = 0$, $i \neq A$, determine the $N - 1$ bilateral rates; wages, prices and expenditure shares follow from the technologies and the demand system Eq. (3.1).

We write $\nu_j = \log e_{Aj}$, with $\nu_A = 0$. The NEER is that of Section 2.4, $d\nu_A^E = \sum_{j \neq A} w_{Aj} \, d\nu_j$ with baseline trade weights $w_{Aj} = (f_{Aj} + f_{jA}) / \sum_{k \neq A} (f_{Ak} + f_{kA})$. The real exchange rate is $q_{Aj} = e_{Aj} P_j / P_A$, with P_i the consumer price index in own currency. A fall in ν_j is an appreciation of A against j and a rise a depreciation; a rise in $\log q_{Aj}$ is a real depreciation of A against j.

The symmetric benchmark is free trade with $\beta_{ij} = 1/(N-1)$, common parameters, and equal endowments and productivities. It retains home bias α_D. Bilateral trade weights then equal $1/(N-1)$, and $m \equiv \alpha_T (1 - \alpha_D)$ denotes the import share of income.

3.2 The two-country benchmark

With $N = 2$ there is a single foreign origin, so ρ is inactive. Linearizing gives $d\mathrm{TB}_A = m M_2 \, d\nu_B + m \rho_D \, d\tau$, so

$$\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho_D}{M_2}, \qquad \rho_D \equiv \underbrace{\alpha_D \eta}_{\text{domestic substitution}} + \underbrace{(1 - \alpha_D)(1 - \alpha_T)}_{\text{rebate / non-tradables}}, \qquad M_2 \equiv 1 + 2\alpha_D(\eta - 1) = \varepsilon_X + \varepsilon_M - 1. \tag{3.4}$$

We call ρ_D the domestic diversion margin. A tariff shifts expenditure toward the domestic economy in two ways: through substitution toward the domestic tradable, and through the income effect of the rebated revenue, part of which is spent on non-tradables. An exchange-rate movement raises no revenue and so has no rebate term, which is why ρ_D differs from M_2. With more than two countries, adjustment also separates into a common movement against all foreign suppliers and relative movements across them.

3.3 The N-country case

The following theorem maps the model's primitives into the coefficients of Proposition 2.4.

Theorem 3.1 (Exchange-rate responses in the symmetric nested-CES model)**.** *At the symmetric benchmark define*

$$M_N \equiv \frac{1}{N-1}\Big[1 + \underbrace{N\alpha_D(\eta-1)}_{\textit{domestic–import adjustment}} + \underbrace{(N-2)\rho}_{\textit{foreign-origin adjustment}}\Big]. \tag{3.5}$$

The eigenvalues of Proposition 2.4 are

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \qquad j_S = \frac{mM_N}{N-1}, \qquad j_D = Nj_S. \tag{3.6}$$

MML holds if and only if $M_N > 0$, and $\eta \geq 1$ suffices. When it holds,

$$\lambda_{\text{avg}} = -\frac{\rho_D}{M_N}, \qquad \lambda_{\text{rel}} = -\frac{\rho}{NM_N}, \qquad d\nu_j = \lambda_{\text{avg}}\bar{t} + \lambda_{\text{rel}}\widetilde{t}_j. \tag{3.7}$$

The eigenvalues are derived in Lemma A.1 and Eq. (A.4), and the proof is in Appendix B. M_N is affine in ρ; we write $M_N \equiv M_N(\rho)$ and suppress the argument except in Appendix C.2, where the dependence on ρ matters.

M_N is the sensitivity of A's aggregate trade balance to its effective exchange rate: $d\text{TB}_A = mM_N\, d\nu_A^E$ at the equal weights of the symmetric benchmark. For $N = 2$ it reduces to M_2. For $N > 2$ it includes ρ, because foreign buyers can switch between A's goods and those of other foreign suppliers.

The equality $j_D = Nj_S$ means that zero-sum relative exchange-rate movements are N times as effective as common movements in closing trade imbalances, which is why λ_{rel} carries the factor $1/N$. A common movement shifts only partners' trade with A, whereas a relative movement also shifts partners' trade with *each other*, and at the symmetric benchmark A accounts for a share $1/N$ of the trade on which relative movements act. Because relative treatment is zero-sum, to first order at the symmetric benchmark it leaves tariff revenue,

income and the tradable/non-tradable split unchanged, so its disturbance is governed by ρ alone.

3.4 A unilateral tariff

Let A raise its tariff on B alone, $t = d\tau\, e_B$. Its mean component is $\bar{t} = d\tau/(N-1)$, B's relative component is $\widetilde{t}_B = \frac{N-2}{N-1} d\tau$, and each bystander's relative component is $\widetilde{t}_c = -\frac{d\tau}{N-1}$.

Proposition 3.2 (Unilateral tariff)**.** *At the symmetric benchmark under MML, to first order,*

$$\frac{d\nu_A^E}{d\tau} = -\frac{\rho_D}{(N-1)M_N} < 0, \qquad \frac{d\nu_B}{d\tau} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} < 0, \qquad \frac{d\nu_c}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N} \tag{3.8}$$

for every untariffed bystander c. Hence:

(i) A's NEER appreciates for every N and ρ.

(ii) A appreciates against the target.

(iii) A depreciates against each untariffed bystander if and only if $\rho > N\rho_D$.

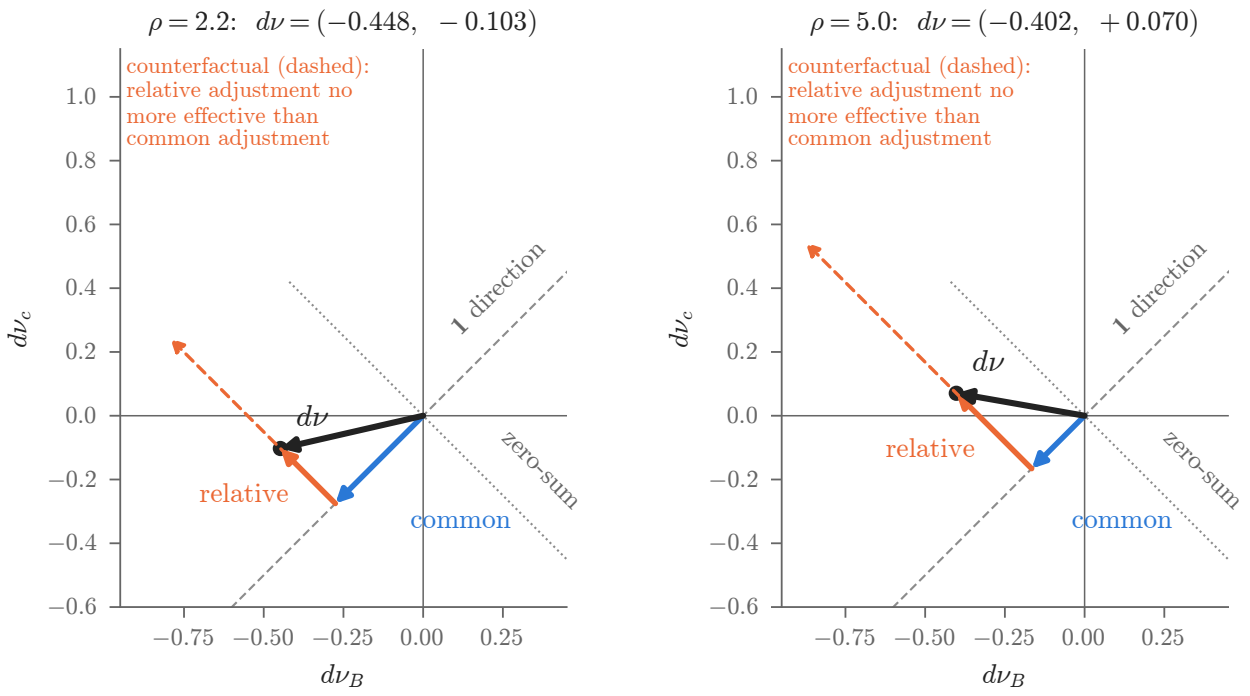
(iv) The same threshold applies when A tariffs any $k \leq N-2$ partners. More generally, for a tariff vector with $\bar{t} > 0$, A depreciates against a partner j receiving a below-average tariff increase, $t_j < \bar{t}$, if and only if $t_j/\bar{t} < 1 - N\rho_D/\rho$, which requires $\rho > N\rho_D$.

Against the target, average protection and relative treatment push in the same direction. Against a bystander they are offsetting pressures: average protection pushes toward an appreciation of A, while the bystander's favorable relative treatment pushes toward a depreciation.

Corollary 3.3 (No reversal with a single elasticity under weak home bias)**.** *Suppose weak home bias, $\alpha_D \geq 1/N$. Then*

$$N\rho_D - \eta = \eta(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \geq 0, \tag{3.9}$$

so $\rho > \eta$ is necessary for a bystander depreciation. In any specification with a single elasticity across all tradable varieties ($\rho = \eta = \sigma$), $\rho - N\rho_D = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)] \leq 0$, so wherever MML holds $d\nu_c/d\tau \leq 0$: A never depreciates against a bystander. The response is strictly negative except at the knife edge $\alpha_D = 1/N$, $\alpha_T = 1$, where it is zero for every $\sigma > 0$. More generally, at $\alpha_D = 1/N$ the response is $-(1-\alpha_T)/(N\sigma)$: if $\alpha_T < 1$ it approaches zero from below as $\sigma \to \infty$, and if $\alpha_T = 1$ it is identically zero.



(a) $\rho = 2.2$: $\rho_D < \rho < 3\rho_D$ **(b)** $\rho = 5$: $\rho > 3\rho_D$

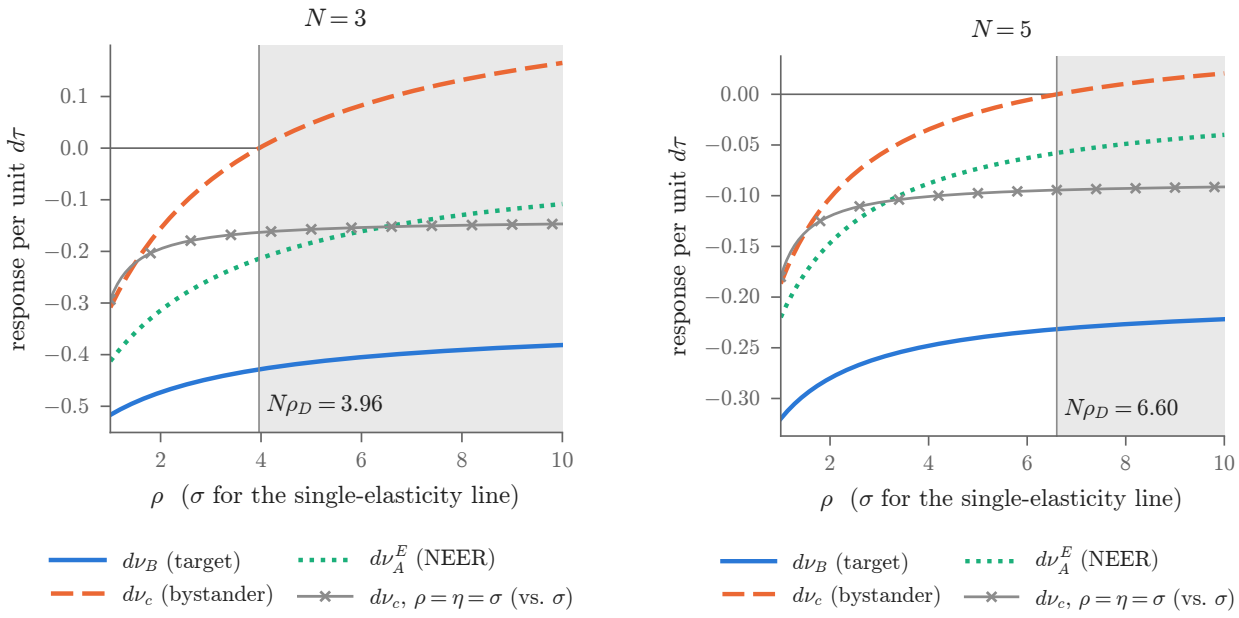
Figure 2: Average protection and relative treatment. Unilateral tariff with $N = 3$: (a) $\rho_D < \rho < 3\rho_D$; (b) $\rho > 3\rho_D$. *Notes:* $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$.

Under weak home bias, the reversal therefore requires cross-origin substitution to exceed home–import substitution. Evidence in Feenstra et al. (2018) and Li et al. (2022) supports allowing ρ and η to differ. At an illustrative US-like calibration ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $\rho_D = 1.32$), the threshold is $N\rho_D = 3.96$ for $N = 3$ and 6.60 for $N = 5$.

At unchanged exchange rates, the trade balance of an individual bystander improves once $\rho > \rho_D$:

$$F_c = \frac{m(\rho - \rho_D)}{(N-1)^2}. \tag{3.10}$$

This is not sufficient for A to depreciate against the bystander in equilibrium. Under MML the exchange-rate response depends on the full vector of trade-balance disturbances, including the target's negative disturbance, and reverses only under the stronger condition $\rho > N\rho_D$. Thus, for $\rho_D < \rho < N\rho_D$, the bystander gains at the initial exchange rates even though A ultimately appreciates against it (Fig. 2a). The gap between the two thresholds reflects the greater effectiveness of relative exchange-rate adjustment identified in Theorem 3.1: if relative and common movements were equally effective, the bystander's equilibrium response would instead change sign when its own disturbance does. Figure 3 plots the three responses against ρ.



(a) $N = 3$ **(b)** $N = 5$

Figure 3: Exchange-rate responses to a unilateral tariff. The bystander response changes sign at $\rho = N\rho_D$. *Notes:* US-like calibration, $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$.

3.5 The role of the bystander bloc

The bystander threshold $N\rho_D$ rises with the number of countries. To see why, note that when the bystanders are symmetric and a tariff shock treats them alike, no relative price within the bystander bloc moves, so the bloc can be collapsed into a single region.

Proposition 3.4 (Three-region representation)**.** *At the symmetric benchmark with $N \geq 3$, consider a tariff shock that treats the $N-2$ bystanders symmetrically and leaves tariffs among them unchanged, and suppose $M_N \neq 0$ (MML is sufficient). To first order, the response of ν_B, the common response of each bystander, and A's NEER are exactly reproduced by a three-region economy consisting of A, B and a composite bystander C with income $N-2$, home share of tradables $h_C = \alpha_D + (1-\alpha_D)\frac{N-3}{N-1}$, and home–import elasticity*

$$\boxed{\eta_C = \frac{(N-3)\rho + 2\alpha_D\,\eta}{N-3+2\alpha_D}.} \tag{3.11}$$

The complete mapping of trade weights and tariff shocks is given in Appendix B.

The equivalence holds for the linearized systems; it is not an aggregation result for finite tariffs.

Trade among the original bystanders becomes the composite's "home" trade, so h_C rises toward one as N grows. In the N-country economy that trade is governed by the cross-origin

elasticity ρ, which is why η_C is a weighted average of η and ρ. As N increases, η_C converges to ρ; it rises with N if $\rho > \eta$, falls if $\rho < \eta$, and is constant if $\rho = \eta$. Under weak home bias the bystander-reversal region $\rho > N\rho_D$ lies inside $\rho > \eta$.

The higher threshold $N\rho_D$ is therefore not simply a consequence of making the bystander bloc larger. As N rises, a greater share of what the composite treats as "home" activity consists of trade among the original bystanders, and that trade adjusts at the cross-origin elasticity ρ. This changes the internal adjustment structure of the bloc and gives relative exchange-rate movements their N-fold effectiveness at the symmetric benchmark (Theorem 3.1). Appendix B shows that holding the composite's internal elasticity fixed at η eliminates this steep N-dependence, confirming that the result reflects the bloc's internal trade structure rather than its size alone.

The nominal bystander reversal need not imply a real reversal because the tariff also raises A's consumer price index. In the low-import-share case, the real reversal therefore requires stronger cross-origin substitution than the nominal reversal; at the US-like calibration with $N = 3$, the nominal threshold is $3\rho_D = 3.96$ and the real threshold is 4.93. The real effective exchange rate continues to appreciate under the conditions given in Appendix C.2, which provides the full derivation and qualifications.

The nested-CES model therefore gives economic content to the decomposition in Section 2. Average protection operates through the domestic diversion margin ρ_D, while relative treatment operates through cross-origin substitution ρ. Under a unilateral tariff, relative treatment can overturn the conventional bilateral result against untariffed partners, but not the effective-rate result at the symmetric benchmark. We next ask how retaliation changes these conclusions.

4 Trade Wars and Preferential Tariffs

We now allow trading partners to respond. Since the analysis is local, simultaneous tariff changes by several countries add up.

4.1 Trade wars

Proposition 4.1 (Retaliation and trade wars)**.** *At the symmetric benchmark with $N \geq 3$ under MML, to first order:*

(i) Under bilateral retaliation (A tariffs B and B tariffs A by $d\tau$), $d\nu_B = 0$ and

$$\frac{d\nu_c}{d\tau} = \frac{\rho - \rho_D}{(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = \frac{N-2}{N-1} \cdot \frac{\rho - \rho_D}{(N-1)M_N}. \tag{4.1}$$

Both belligerents depreciate against every bystander, and in effective terms, if and only if $\rho > \rho_D$. Cross rates among bystanders are unchanged.

(ii) Under a multilateral trade war (A tariffs every partner, and every partner retaliates against A), $d\nu_j = d\nu_A^E = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N}\, d\tau$ for all j, so A depreciates against every partner if and only if $\rho > \rho_D$.

B's retaliation adds a second relative-treatment effect in the bystander's favor, which offsets the $1/N$ dilution. The bystander threshold therefore falls from $N\rho_D$ under a unilateral tariff to ρ_D, the point at which the bystander's trade balance improves at unchanged exchange rates. At the US-like calibration, $\rho_D = 1.32$, below the illustrative range $\rho \in [2, 4]$ shaded in Fig. 4. Estimates in Broda and Weinstein (2006) and Feenstra et al. (2018) place median cross-foreign elasticities around 3–4. For values in this range the model predicts that retaliation depreciates the tariffing country's currency against bystanders and in effective terms while leaving the belligerents' bilateral rate unchanged. The model can thus reproduce the pattern documented by Ostry et al. (2025), including the near-stability of the dollar against the renminbi in tit-for-tat episodes, with flexible prices, balanced trade and no financial channel.

The two experiments differ in where adjustment falls. Under bilateral retaliation the belligerents' rate is fixed and adjustment runs against bystanders. In a multilateral war there are no bystanders, and all of A's rates move together.

4.2 Preferential tariffs and competing exporters

In our model class the bystander's exchange rate is its terms of trade with A, $\mathrm{ToT}_{cA} = e_{Ac}$, so the response in Eq. (3.8) is the third-country terms-of-trade effect of a discriminatory tariff, the central object of the preferential-trade-agreement literature since Viner (1950) and Mundell (1964). In the canonical competing-exporter model, an importer A that does not produce the good buys it from B and C under quasi-linear preferences. With q_j exporter j's price, E_j its export supply and M_A the importer's demand, a tariff on B always improves C's terms of trade,

$$\frac{dq_C}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} > 0.$$

In our model the bystander gains only if $\rho > N\rho_D$. The difference is that A also produces a competing tradable: part of the demand diverted from B goes to A's own good, along with the rebate's income effect, so the third-country effect survives only when cross-origin substitution is strong relative to domestic diversion. Appendix C.1 shows that at a positive common external tariff the threshold rises, and that a discrete preferential cut requires a

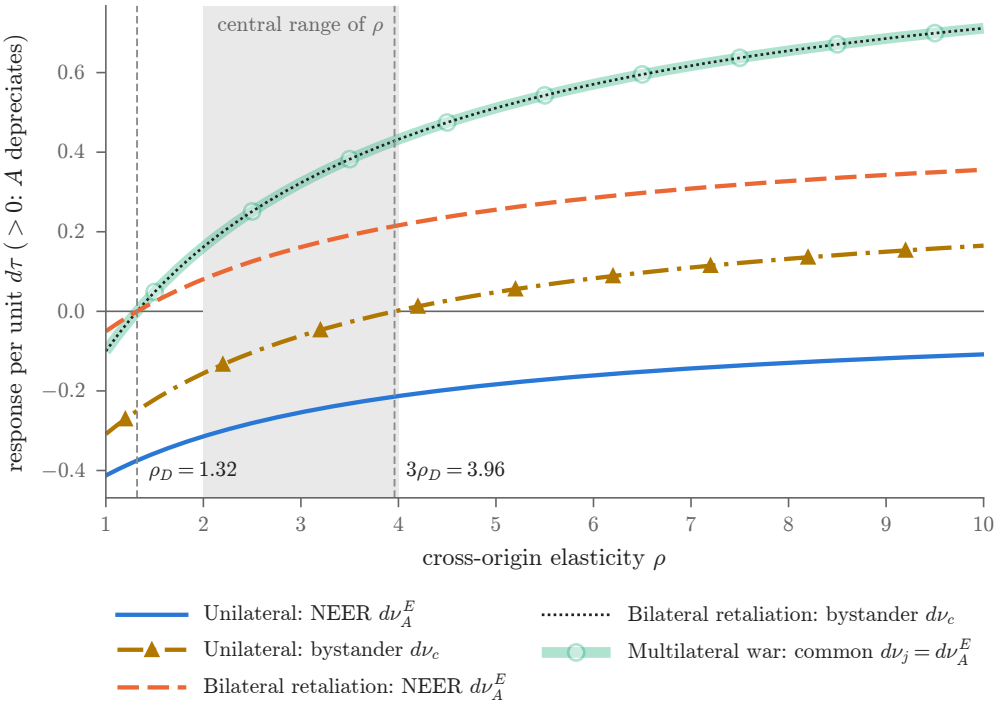


Figure 4: Exchange-rate responses across tariff configurations. Unilateral tariffs, bilateral retaliation, and a multilateral trade war for $N = 3$. Retaliation lowers the bystander threshold from $3\rho_D$ to ρ_D. *Notes:* US-like calibration; shading denotes $\rho \in [2, 4]$.

higher ρ than the local threshold suggests.

5 Asymmetry

Under partner symmetry the common and zero-sum subspaces are eigenspaces of the adjustment matrix (Proposition 2.4), so average protection moves the effective exchange rate and relative treatment moves bilateral rates around it. Asymmetry in country sizes, origin weights or home bias breaks this structure. This section asks which of the symmetric results survive. Two weighted averages of bilateral responses must be kept apart: the trade-weighted NEER of Section 2.4, with weights w_{Aj} that combine A's imports from and exports to j, and the aggregate terms of trade, which weight the same responses by A's import shares b_{Aj}. The two coincide at the symmetric benchmark and differ away from it, and the second is the welfare-relevant object.

5.1 The aggregate terms of trade and welfare

Let b_{Aj} denote partner j's share in A's baseline import expenditure, so that $\sum_{j\neq A} b_{Aj} = 1$, and let $R_j^B \equiv \partial \nu_j / \partial \tau_{AB}$ be the response of A's bilateral rate against j to a marginal tariff by A on B; $R_j^B > 0$ is a depreciation of A against j. Define

$$R_{A|B} \equiv \sum_{j\neq A} b_{Aj} R_j^B. \tag{5.1}$$

Since each producer's price is fixed in its own currency, the log change in A's import-weighted aggregate terms of trade is $-\sum_{j\neq A} b_{Aj}\, d\nu_j$, and a marginal tariff on B improves it if and only if $R_{A|B} < 0$.

The welfare result is stated at a balanced free-trade baseline of the model of Section 3.1 with interior shares, lump-sum rebate of tariff revenue, and exchange rates differentiable in tariffs, which holds whenever J is nonsingular.

Lemma 5.1 (First-order welfare)**.** *Under these assumptions, let $d\tau$ be any marginal tariff vector, imposed by A or by other countries. To first order, the change in the welfare of A's representative household, in units of A's income, is*

$$\frac{dU_A}{U_A' I_A} = -\, m_A \sum_{j\neq A} b_{Aj}\, d\nu_j, \tag{5.2}$$

where m_A is A's import share of income. A marginal tariff change therefore raises A's own welfare to first order if and only if it improves A's import-weighted aggregate terms of trade.

At free trade, the first-order tariff-revenue gain offsets the direct increase in import prices, leaving the welfare effect of the induced exchange-rate adjustment. The formal proof is in Appendix B.

Because the first-order welfare effect is linear in the tariff vector, write, for tariffs imposed by A,

$$\frac{dU_A}{U_A' I_A} = \sum_{j\neq A} g_j\, d\tau_{Aj}, \qquad g_j \equiv -m_A\, R_{A|j}, \tag{5.3}$$

where $R_{A|j}$ is the response Eq. (5.1) to a tariff on j alone. The component g_j is the first-order welfare gain to A from a marginal tariff on j, and a marginal uniform tariff, $d\tau_{Aj} = dt$ for all $j \neq A$, is worth $\sum_j g_j\, dt$.

Proposition 5.2 (Uniform tariffs)**.** *Consider any balanced free-trade baseline at which MML and the assumptions of Lemma 5.1 hold; country sizes, origin shares, home biases and parameters may differ across countries. Let A impose a marginal uniform tariff on all foreign partners, $d\tau_{Aj} = dt > 0$ for all $j \neq A$, with all other tariffs unchanged. Then:*

(i) the tariff disturbs every partner's trade balance by

$$F_j = -f_{Aj}\,\rho_D^A\,dt, \qquad \rho_D^A \equiv \eta_A h_A + (1-h_A)(1-\alpha_T^A) > 0, \tag{5.4}$$

where h_A is A's home share of tradable spending and η_A, α_T^A are A's own parameters;

(ii) A's currency appreciates weakly against every partner, $d\nu_j \le 0$, and strictly against every partner if A imports from all of them or J is irreducible;

(iii) A's import-weighted aggregate terms of trade improve and A's own welfare rises to first order: $\sum_{j\neq A} g_j > 0$.

A uniform tariff leaves relative import prices across foreign suppliers unchanged, so cross-origin substitution creates no differential incidence across partners, and the exchange-rate adjustment preserves the conventional aggregate result. The result is local, concerns A's own welfare at a balanced free-trade baseline, and establishes nothing about finite tariffs, retaliation, global welfare, or a globally optimal tariff. It is a local, own-country counterpart of the classical terms-of-trade argument for a small tariff (Johnson, 1953; Broda et al., 2008; Costinot et al., 2015); as in Bagwell and Staiger (1999, 2002), nondiscrimination confines a country's tariffs to a single terms-of-trade channel. The proof is in Appendix B.

A uniform tariff restricts the sum of the partner-specific welfare gradients, not every component separately; a tariff on one partner can therefore still lower A's welfare. With $N = 3$, $g_B + g_C > 0$, so $g_B < 0$ implies $g_C > 0$.

5.2 The aggregate terms-of-trade condition

Separate the target from the remaining suppliers. Define

$$\bar{R}_{-B}^B \equiv \frac{\sum_{j\neq A,B} b_{Aj} R_j^B}{1-b_{AB}},$$

the import-weighted average response of A's rates against all bystanders, each bystander weighted by its share in A's imports from suppliers other than B. Then $R_{A|B} = b_{AB}R_B^B + (1-b_{AB})\bar{R}_{-B}^B$: A's exposure to the target, its import weight on B times the move against B, plus its exposure to the bystanders.

Definition 5.3 (Aggregate terms-of-trade condition)**.** Unilateral tariffs by A have the conventional aggregate terms-of-trade effect if, for every target B,

$$b_{AB}R_B^B + (1-b_{AB})\bar{R}_{-B}^B < 0. \tag{5.5}$$

Suppose A appreciates against the target, $R_B^B < 0$, and depreciates on average against the bystanders, $\bar{R}_{-B}^B > 0$. The condition then fails if and only if

$$\frac{\bar{R}_{-B}^B}{-R_B^B} > \frac{b_{AB}}{1 - b_{AB}}. \tag{5.6}$$

The left side is the *equilibrium incidence*: the average adverse move against bystanders per unit of favorable move against the target. The right side is *relative import exposure*: A's import expenditure on the target relative to its import expenditure on all other suppliers. Both are positive pure numbers. Given $R_B^B < 0$, a bilateral reversal against at least one bystander is necessary but not sufficient for an aggregate reversal: the adverse moves must also be large enough and carry enough weight in A's import basket.

In terms of Eq. (5.3), the condition requires every partner-specific component g_B to be positive, whereas Proposition 5.2 restricts only their sum. The condition has a three-country algebraic form for any configuration of bystanders, because $\bar{R}_{-B}^B$ is defined as their import-weighted average; this does not make a heterogeneous bloc a literal representative country (Section 5.4).

5.3 Asymmetric partners in the three-country model

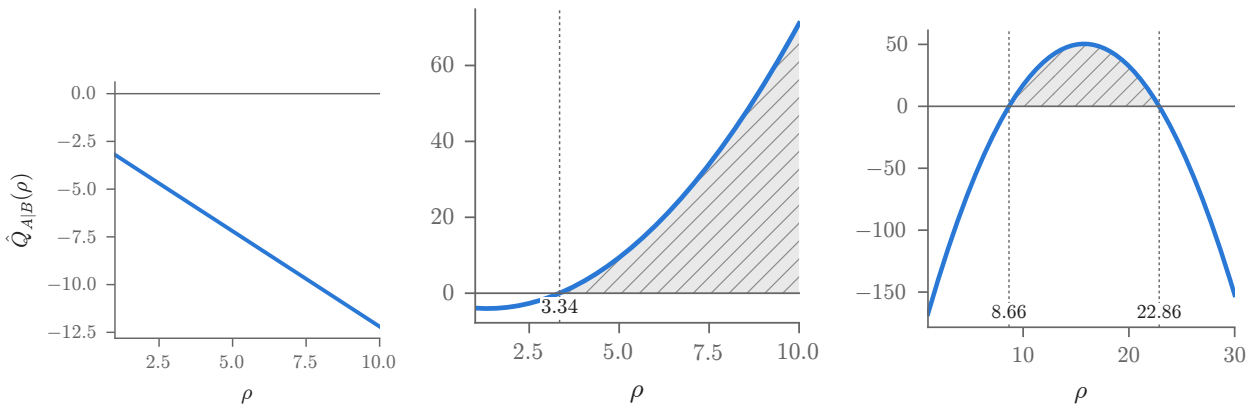
Fix a balanced free-trade baseline of the three-country model, that is, the import-origin shares b_{ij}, the home shares h_i, the incomes y_i and all parameters other than ρ, and let only ρ vary. Write the local system as $A(\rho)R = F(\rho)$, with $A \equiv -J$ and $R = (R_B^B, R_C^B)'$. Since ρ enters the linearized share responses (Eq. (A.1)) only through the factor $(1-\rho)$, every entry of A and F is affine in ρ, and so is the adjugate of the 2×2 matrix A. Hence

$$R_{A|B}(\rho) = \frac{b' \operatorname{adj}(A(\rho))F(\rho)}{\det A(\rho)}, \qquad b' \operatorname{adj}(A(\rho))F(\rho) = q_2\rho^2 + q_1\rho + q_0 \equiv Q_{A|B}(\rho). \tag{5.7}$$

Under MML $\det A(\rho) > 0$, so at every ρ at which MML holds at the fixed baseline, $R_{A|B}$ has the sign of $Q_{A|B}(\rho)$ and the aggregate condition is $Q_{A|B}(\rho) < 0$. The bilateral bystander response has the same structure; with a common α_T, its numerator has leading coefficient $\alpha_T^2\, b_{AB}b_{AC}b_{CA}b_{CB}(1-h_A)(1-h_C)y_Ay_C > 0$ (Appendix C.3).

Proposition 5.4 (Three cases)**.** *Fix a balanced free-trade baseline of the three-country model as above, and consider values $\rho > 0$ at which MML holds at that baseline. The following three cases are those used in this paper; they are not an exhaustive classification of the signs of (q_2, q_1, q_0).*

(a) Symmetric benchmark: $q_2 = 0$ and $Q_{A|B}(\rho) < 0$ for all ρ. The aggregate condition



(a) Symmetric benchmark **(b)** Small, low-weight target **(c)** Asymmetric network

Figure 5: Aggregate terms-of-trade effects under asymmetry. Examples with no reversal, one threshold, and two thresholds. Shading denotes regions in which a marginal discriminatory tariff worsens the aggregate terms of trade.

always holds.

(b) $q_2 > 0$ *and* $q_0 < 0$*:* $Q_{A|B}$ *has exactly one positive root* ρ^**. The condition holds for* $\rho < \rho^*$ *and fails for* $\rho > \rho^*$*.*

(c) $q_2 < 0$*,* $q_1 > 0$*,* $q_0 < 0$ *and a positive discriminant:* $Q_{A|B}$ *has two positive roots* $\rho_1 < \rho_2$*. The condition fails only for* $\rho_1 < \rho < \rho_2$*.*

Case (b) is the economically central asymmetric case: the conventional aggregate effect holds at low cross-origin substitutability and is reversed above a single threshold. Case (c) requires richer network asymmetry and is reported as a robustness example. The three panels of Fig. 5 are parameterized examples of the three cases (Appendix C.3).

Aggregate terms-of-trade reversal is more likely when the targeted partner is small relative to untariffed suppliers, when it has a low weight in A's imports, and when foreign suppliers are sufficiently close substitutes. The appreciation against a small or lightly weighted target receives little weight in A's import price index, the adverse movements against larger bystanders receive more, and strong cross-origin substitution redirects expenditure toward those bystanders. In the local model, at a balanced free-trade baseline satisfying MML, a discriminatory marginal tariff can therefore reduce A's own welfare to first order, whereas by Proposition 5.2 a marginal uniform tariff at the same baseline raises it. Appendix C.3 reports the numerical examples.

5.4 Effective exchange rates and heterogeneous bystanders

The NEER uses trade weights w_{Aj}, which include A's exports to j, whereas the aggregate terms of trade use import weights b_{Aj}. Under asymmetry the two weighting schemes differ, so the NEER threshold can differ from the aggregate terms-of-trade threshold; when the target buys much more from A than A buys from it, the favorable move against the target counts for more in the NEER. Appendix C.3 reports a numerical illustration.

For $N \geq 4$, heterogeneous bystanders generally cannot be collapsed into the exact representative bystander of Proposition 3.4. The bloc's average response depends on its internal adjustment structure, including the block J_{RR} of bystanders' responses to their own rates, their responses J_{RB} to the rate against the target, and their disturbances F_R (Appendix C.3), even though the identity $R_{A|B} = b_{AB}R_B^B + (1 - b_{AB})\bar{R}_{-B}^B$ always holds. The three-country structure is therefore a diagnostic of sign mechanisms rather than a quantitative representation of a heterogeneous network.

5.5 A stylized reading of the 2025 episode

The 2025 exercise is neither a calibration nor a causal test. The model is static and has flexible prices, its central analysis assumes balanced trade, and observed exchange rates are high-frequency asset prices. We use the model only to classify the signs associated with different tariff configurations and do not attempt to match magnitudes. A movement is *consistent with* the trade channel when the model predicts its sign in the configuration that best approximates the phase, and *not explained by* the trade channel when the model predicts the opposite sign for every admissible ρ; sign consistency is not causal identification. Table 1 and Fig. 6 report the data, with a positive number a depreciation of the dollar; the Federal Reserve's broad dollar index is the closest counterpart of the model's trade-weighted NEER, and the index excluding China isolates the bystanders.

Phase 1: China-focused escalation (February 1–April 2). The United States raised its tariff on China by 20 percentage points in two steps, and China responded only with product-specific tariffs. At the same time the United States imposed tariffs on Canada and Mexico and Canada retaliated, so the phase is not a clean unilateral experiment. The dollar was roughly unchanged against the renminbi (−0.3 percent on daily data, +0.6 percent on monthly averages), which does not contradict the unilateral sign restriction, and weakened against bystanders, by 4.5 percent against the euro and 1.9 percent against the broad index excluding China on daily data. If the phase is approximated by a unilateral tariff on China, this weakness is consistent with the high-ρ bystander mechanism of Proposition 3.2.

Table 1: Exchange-rate changes within the three phases of the 2025 episode (percent; positive = dollar depreciation)

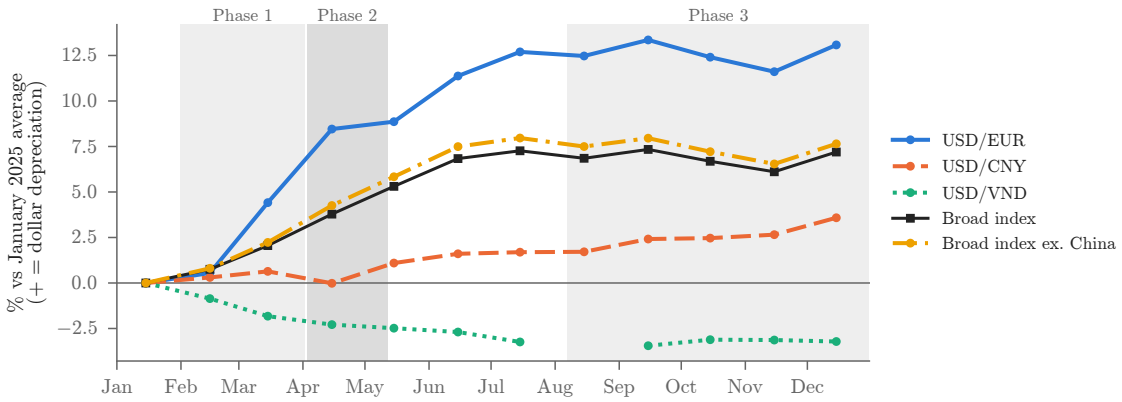
	Phase 1 Feb 1–Apr 2		Phase 2 Apr 3–May 12		Phase 3 Aug 7–Dec 31	
	Monthly	Daily	Monthly	Daily	Monthly	Daily
EUR	4.4	4.5	3.9	2.2	0.3	0.8
CNY	0.6	−0.3	−0.7	0.9	1.9	2.7
CAD	0.2	1.0	2.7	2.2	−0.7	0.2
MXN	1.6	0.6	1.0	4.5	3.4	3.5
VND	−1.8	-2.1^a	−0.5	-0.5^a	0.0	0.5^a
Broad index	2.1	1.7	1.7	2.3	−0.1	0.6
Broad index ex. China	2.2	1.9	2.0	2.5	−0.3	0.4

Notes: Positive values denote dollar depreciation. Monthly columns compare monthly averages (January with March, March with April, July with December); daily columns compare the last available observations at the phase boundaries (January 31 to April 2, April 2 to May 12, August 6 to December 31). VND daily entries (marked ^{a}) use BIS end-of-month rates. Sources: Federal Reserve H.10 and BIS. See Appendix C.5 for construction and data details.

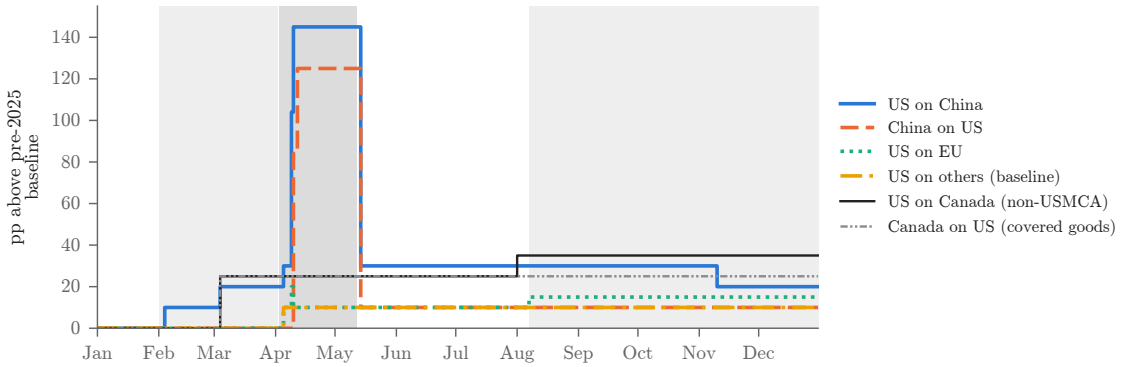
The 1.7 percent fall of the dollar against the full broad index cannot separate unilateral from retaliation effects and may also reflect other contemporaneous shocks, so the data do not identify the trade channel.

Phase 2: US–China retaliation (April 3–May 12). The United States and China raised tariffs on each other in a mirror escalation, to 145 and 125 percent, while other partners faced a common 10 percent baseline tariff. The phase is therefore close to the bilateral-retaliation experiment of Proposition 4.1 with a near-uniform component added. For bilateral retaliation the model predicts a nearly stable bilateral rate between the belligerents, dollar weakness against bystanders, and effective depreciation when $\rho > \rho_D$. The phase displayed this pattern: the dollar moved little against the renminbi (+0.9 percent on daily data, −0.7 percent on monthly averages) but fell by 2.2–3.9 percent against the euro, 2.0–2.5 percent against the broad index excluding China and 1.7–2.3 percent against the full broad index. The model can thus reproduce the configuration documented by Ostry et al. (2025). As an interpretation outside the static model, markets may price the tariff configuration expected to persist, in which case expected retaliation would push in the same direction.

Two features of the configuration push in opposite directions. High cross-origin substitutability, suggested by the rapid reallocation of trade between China and other suppliers in the earlier US–China trade war (Fajgelbaum et al., 2024; Freund et al., 2024; Alfaro and Chor, 2023), pushes toward dollar weakness against bystanders. China's size and import weight push against the depreciation mechanism: a larger and more heavily weighted tar-



(a) Exchange rates, percent deviation from the January 2025 average (positive = dollar depreciation)



(b) Headline tariff increases above the pre-2025 baseline (percentage points)

Figure 6: Exchange rates and headline tariff increases during 2025. Panel (a) reports exchange-rate movements relative to January 2025; panel (b) reports tariff increases above the pre-2025 baseline. Shading denotes the three policy phases. Sources: Federal Reserve, BIS, Federal Register, Department of Finance Canada, and China's State Council Tariff Commission.

get raises the bystander threshold and moves the aggregate terms-of-trade condition toward its conventional sign. In the illustrative mapping of Appendix C.4, the bilateral bystander threshold for a US tariff on China is about 3.9–4.0 and the aggregate terms-of-trade threshold about 4.5–4.7. A quantitative network model is needed to determine the net sign and the magnitudes.

Phase 3: Near-uniform tariffs with residual retaliation (August 7–December 31). After the Geneva truce and the August framework rates, US tariffs became more nearly uniform across partners, which raised the average-protection component, while residual gaps against China and Canada, met by retaliation, remained. In the model the average-protection component pushes toward effective dollar appreciation (Propositions 5.2 and 2.4), whereas the residual gaps can push toward depreciation if $\rho > \rho_D$. Within the phase the

broad index was nearly flat (−0.1 percent on monthly averages, +0.6 percent on daily data), with no corresponding broad effective appreciation. Over the year as a whole, the dollar was 7.2 percent weaker against the broad index and 7.6 percent weaker against the index excluding China in December than in January, on monthly averages, most of it accumulated before August. The static trade channel alone does not explain the full effective depreciation.[1]

A quantitative assessment would require the retaliation structure, cross-origin substitution elasticities, partner size and trade weights, the degree to which protection is uniform or discriminatory, and a dynamic mechanism linking policy expectations to asset prices.

6 Conclusion

With two countries, a tariff moves a single exchange rate, and the bilateral rate, the effective rate and the terms of trade coincide. With three or more countries they separate. The general framework writes every bilateral response as a multilateral adjustment matrix applied to the vector of tariff-induced trade-balance disturbances. The multilateral Marshall–Lerner condition makes that adjustment stable but does not determine the sign of every bilateral response (Section 2).

Under partner symmetry, any tariff vector separates into average protection, which moves the effective rate, and relative treatment, which moves bilateral rates around it (Proposition 2.4). The nested-CES model maps these components into the domestic diversion margin ρ_D and cross-origin substitution ρ. Locally, at the symmetric balanced free-trade benchmark and under MML, a unilateral tariff appreciates the tariffing country's effective rate but depreciates its currency against untariffed bystanders when $\rho > N\rho_D$ (Proposition 3.2); retaliation lowers the relevant threshold to ρ_D (Proposition 4.1).

Under strong asymmetry the reversal can extend to the import-weighted aggregate terms of trade and to own welfare: a discriminatory marginal tariff can lower the tariffing country's welfare to first order when the target is small, carries a low weight in its imports, and foreign suppliers are close substitutes (Sections 5.2 and 5.3). Uniform protection, by contrast, retains the conventional local result under the conditions of Proposition 5.2.

Applied to 2025, the framework classifies tariff configurations by the exchange-rate signs they imply; it does not identify causes (Section 5.5). A quantitative multi-country network model, with measured elasticities, trade weights and retaliation structure, is the natural next step.

[1] Vietnam illustrates why classification matters: the dollar strengthened against the dong by 3.2 percent over the year, but Vietnam moved during 2025 from a bystander-like position to a direct target (Appendix C.5).

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A The Linearized System

This appendix derives the linear system Eq. (2.2) of the nested-CES model of Section 3.1 and evaluates it at the symmetric benchmark.

A.1 General linearization

All derivatives are first-order and evaluated at a balanced free-trade baseline, $\tau_{ij} = 0$ and $\mathrm{TB}_i = 0$ for every $i \in \mathcal{I}$. Values are in the common unit of account of Section 2.1, with $v_A = 1$. Write $\chi_l = \log v_l$ for $l \in \mathcal{I}$; then $\nu_j = \log e_{Aj} = \chi_j - \chi_A$, $\nu_A \equiv 0$, and $d\log e_{ij} = d\nu_j - d\nu_i$.

The baseline is described by origin shares b_{ij} within i's import bundle ($b_{ii} = 0$, $\sum_{j\neq i} b_{ij} = 1$), home shares h_i of i's tradable spending, and incomes $y_i \equiv \tilde{I}_i$ in the common unit. The share of i's income spent on j's good is $s_{ij} = \alpha_T^i(1-h_i)b_{ij}$, and the border value of i's imports from j is $f_{ij} = s_{ij}y_i$; η_i, ρ_i and α_T^i may differ by country. Log-differentiating the price indices of Section 3.1, the demand system Eq. (3.1), income Eq. (3.2) and the trade balance Eq. (3.3) gives

$$
\begin{aligned}
dp_{ij} &= d\nu_j - d\nu_i + d\tau_{ij}, \qquad dp_{ii} = 0,\\
d\log P_{Mi} &= \textstyle\sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{Ti} = (1-h_i)\, d\log P_{Mi},\\
d\log s_{ii} &= -(1-\eta_i)(1-h_i)\, d\log P_{Mi},\\
d\log s_{ij} &= (1-\rho_i)(dp_{ij} - d\log P_{Mi}) + (1-\eta_i)h_i\, d\log P_{Mi},\\
d\log \tilde{I}_i &= d\nu_i + \textstyle\sum_{j\neq i} s_{ij}\, d\tau_{ij},\\
d\mathrm{TB}_i &= \textstyle\sum_{k\neq i} f_{ki}(d\log s_{ki} + d\log \tilde{I}_k - d\tau_{ki}) - \sum_{j\neq i} f_{ij}(d\log s_{ij} + d\log \tilde{I}_i - d\tau_{ij}),
\end{aligned}
\tag{A.1}
$$

where dp_{ij} is the log change in i's consumer price of j's good in i's currency. At free trade only the direct revenue term survives in $d\log \tilde{I}_i$, and $d\log f_{ij} = d\log s_{ij} + d\log \tilde{I}_i - d\tau_{ij}$. Stacking $d\mathrm{TB}_i$ for $i \in \mathcal{I}_{-A}$ in the order of Eq. (2.2) gives J and G, and hence $G^{(A)}$, at any balanced free-trade baseline. The $N \times N$ Jacobian $\widetilde{J}$ with respect to $(\chi_l)_{l\in\mathcal{I}}$ of Section 2.3 follows with each $d\nu_i$ replaced by $d\chi_i$.

A.2 The symmetric benchmark

At the symmetric benchmark the country subscripts on η, ρ and α_T are dropped; $b_{ij} = 1/(N-1)$, $h_i = \alpha_D$, $y_i = 1$, and $f_{ij} = s_{ij} = m/(N-1)$ with $m = \alpha_T(1-\alpha_D)$.

Lemma A.1 (Jacobian at the symmetric baseline)**.** *At the symmetric balanced baseline, $\widetilde{J}_{il} = mM_N/(N-1)$ for $i, l \in \mathcal{I}$ with $l \neq i$, and $\widetilde{J}_{ii} = -mM_N$. The reduced matrix*

$J = (\widetilde{J}_{il})_{i,l\in\mathcal{I}_{-A}}$ *satisfies*

$$-J = j_S\,(NI - \mathbf{1}\mathbf{1}'), \qquad j_S = \frac{mM_N}{N-1}, \qquad j_D = Nj_S. \tag{A.2}$$

Proof. Set $d\tau = 0$ in the $\widetilde{J}$ version of Eq. (A.1), so $dp_{ij} = d\chi_j - d\chi_i$ and $d\log\tilde{I}_i = d\chi_i$. With $\Sigma \equiv \sum_{j\in\mathcal{I}} d\chi_j$,

$$d\log P_{Mk} = \frac{\Sigma - N\,d\chi_k}{N-1}, \qquad \frac{\partial \log P_{Mk}}{\partial \chi_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

With equal weights, $\sum_{j\neq i}(dp_{ij} - d\log P_{Mi}) = 0$, so $\sum_{j\neq i} d\log s_{ij} = (N-1)(1-\eta)\alpha_D\, d\log P_{Mi}$ and

$$d\mathrm{TB}_i = \frac{m}{N-1}\Big[\sum_{k\neq i}(d\log s_{ki} + d\chi_k) - (N-1)(1-\eta)\alpha_D\, d\log P_{Mi} - (N-1)\, d\chi_i\Big].$$

Fix $l \neq i$. In $\sum_{k\neq i}\log s_{ki}$, the term $k = l$ contributes $-(1-\eta)\alpha_D$ and each of the $N-2$ terms $k \notin \{i,l\}$ contributes $[-(1-\rho) + (1-\eta)\alpha_D]/(N-1)$, so

$$\frac{\partial}{\partial\chi_l}\sum_{k\neq i}\log s_{ki} = -(1-\rho)\frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import price-index term $-(1-\eta)\alpha_D$, and the import income term zero. Using $(N-1) - (1-\rho)(N-2) = 1 + (N-2)\rho$,

$$\widetilde{J}_{il} = \frac{m}{(N-1)^2}\Big[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1)\Big] = \frac{mM_N}{N-1}.$$

The trade balance is homogeneous of degree one in currency values and zero at the baseline, so the row sums of $\widetilde{J}$ vanish and $\widetilde{J}_{ii} = -(N-1)\widetilde{J}_{il} = -mM_N$. Deleting row and column A gives diagonal entries $-j_S(N-1)$ and off-diagonal entries j_S, which is Eq. (A.2). Since $\mathbf{1}'\mathbf{1} = N-1$, $-J$ has eigenvalue j_S on $\mathrm{span}\{\mathbf{1}\}$ and Nj_S on the zero-sum subspace. □

At $d\nu = 0$, a change $t = (t_j)_{j\in\mathcal{I}_{-A}}$, $t_j = d\tau_{Aj}$, in A's tariffs moves only A's import prices and income, so among A's partners only i's exports to A change:

$$d\mathrm{TB}_i\Big|_{\nu} = f_{Ai}\Big(d\log s_{Ai} + d\log\tilde{I}_A - t_i\Big) = \frac{m}{N-1}\Big[-\rho_D\,\bar{t} - \rho\,(t_i - \bar{t})\Big], \tag{A.3}$$

using $d\log P_{MA} = \bar{t}$, $d\log s_{Ai} = (1-\rho)(t_i - \bar{t}) + (1-\eta)\alpha_D\bar{t}$, $d\log\tilde{I}_A = m\bar{t}$ and $1 - (1-\eta)\alpha_D - m = \rho_D$. Equivalently, $G^{(A)} = \frac{m}{N-1}\Big[-\rho I + \frac{\rho-\rho_D}{N-1}\mathbf{1}\mathbf{1}'\Big]$, with eigenvalues

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \tag{A.4}$$

on span$\{\mathbf{1}\}$ and on the zero-sum subspace.

A.3 The two-country case

At $N = 2$ the margin $(1-\rho)$ is inactive, $dp_{AB} = d\log P_{MA} = d\nu_B + d\tau$ and $dp_{BA} = d\log P_{MB} = -d\nu_B$, so

$$d\log s_{AB} = (1-\eta)\alpha_D(d\nu_B + d\tau), \qquad d\log s_{BA} = -(1-\eta)\alpha_D\, d\nu_B,$$
$$d\log \tilde{I}_A = m\, d\tau, \qquad d\log \tilde{I}_B = d\nu_B.$$

Substituting into $\mathrm{TB}_A = f_{BA} - f_{AB}$,

$$d\mathrm{TB}_A = m\Big[\big(1 + 2\alpha_D(\eta-1)\big)d\nu_B + \big(1 - m + \alpha_D(\eta-1)\big)d\tau\Big] = mM_2\, d\nu_B + m\rho_D\, d\tau,$$

with $M_2 = 1 + 2\alpha_D(\eta-1)$ and $\rho_D = 1 - m + \alpha_D(\eta-1) = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T)$. Setting $d\mathrm{TB}_A = 0$ gives Eq. (3.4).

For the elasticity form, let Q_M be A's import volume, with consumer price $p = p_{AB} = (1+\tau)e_{AB}$, and Q_X its export volume, priced at $1/e_{AB}$ in B's currency. At $\tau = 0$ incomes are fixed in own currency when ν moves, and import spending is $s_{AB}I_A = pQ_M$, so at fixed τ

$$\varepsilon_M \equiv -\frac{\partial \log Q_M}{\partial \nu} = -\frac{\partial \log Q_M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1 + \alpha_D(\eta-1),$$

and by symmetry $\varepsilon_X \equiv \partial \log Q_X/\partial\nu = 1 + \alpha_D(\eta-1)$. With $\mathrm{TB}_A = Q_X - e^{\nu}Q_M$, $\mathrm{TB}_{A\nu} = X(\varepsilon_X + \varepsilon_M - 1)$ (Section 2.1), so $\varepsilon_X + \varepsilon_M - 1 = M_2$.

A.4 Unilateral and discriminatory tariffs

Here $N \geq 3$, the baseline is the symmetric benchmark, and $M_N \neq 0$; sign statements additionally use MML, $M_N > 0$.

For a unilateral tariff, $t = d\tau\, e_B$, and an individual bystander c, $\bar{t} = d\tau/(N-1)$, $\widetilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\widetilde{t}_c = -\frac{d\tau}{N-1}$. By Eq. (3.7),

$$d\nu_B = \frac{\lambda_{\mathrm{avg}} + (N-2)\lambda_{\mathrm{rel}}}{N-1}\, d\tau = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\, d\tau,$$
$$d\nu_c = \frac{\lambda_{\mathrm{avg}} - \lambda_{\mathrm{rel}}}{N-1}\, d\tau = \frac{\rho - N\rho_D}{N(N-1)M_N}\, d\tau,$$

while $d\nu_A^E = \lambda_{\mathrm{avg}}\, d\tau/(N-1) = -\rho_D\, d\tau/[(N-1)M_N]$ and $d\log e_{Bc} = d\nu_c - d\nu_B = -\lambda_{\mathrm{rel}}\, d\tau = \rho\, d\tau/(NM_N)$, as in Eq. (3.8). By Eq. (A.3), the target's and each bystander's disturbances

per unit $d\tau$ are

$$F_B = -\frac{m[\rho_D + (N-2)\rho]}{(N-1)^2} < 0, \qquad F_c = \frac{m}{N-1}\Big[-\frac{\rho_D}{N-1} + \frac{\rho}{N-1}\Big] = \frac{m(\rho-\rho_D)}{(N-1)^2},$$

the second of which is Eq. (3.10).

For the single-elasticity case, $N\rho_D = N[1 + \alpha_D(\eta-1) - \alpha_T(1-\alpha_D)]$ rearranges to Eq. (3.9). With $\rho = \eta = \sigma$, $d\nu_c/d\tau = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)]/[N(N-1)M_N]$. At $\alpha_D = 1/N$, $N\rho_D - \sigma = (N-1)(1-\alpha_T)$ and $M_N = \sigma$, so $d\nu_c/d\tau = -(1-\alpha_T)/(N\sigma)$.

For a general tariff vector with $\bar{t} > 0$, substituting λ_{avg} and λ_{rel} into $d\nu_j = \lambda_{\text{avg}}\bar{t} + \lambda_{\text{rel}}(t_j - \bar{t})$ shows that, under MML, A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_D\,\bar{t} \qquad \Longleftrightarrow \qquad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_D}{\rho}. \tag{A.5}$$

For a tariff increase on j, $t_j \geq 0$, the left side of the second inequality is nonnegative, so depreciation against j requires both $\rho > N\rho_D$ and a tariff differential large enough that $t_j/\bar{t}$ falls below $1 - N\rho_D/\rho$.

If A raises its tariff by $d\tau$ on k partners, $1 \leq k \leq N-1$, then $\bar{t} = k\,d\tau/(N-1)$, $\widetilde{t}_B = (N-1-k)\,d\tau/(N-1)$ for every targeted B and $\widetilde{t}_c = -k\,d\tau/(N-1)$ for every untargeted c, and Eq. (3.7) gives

$$\frac{d\log e_{Ac}}{d\tau} = \frac{k(\rho - N\rho_D)}{N(N-1)M_N}, \qquad \frac{d\log e_{AB}}{d\tau} = -\frac{Nk\rho_D + (N-1-k)\rho}{N(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = -\frac{k\rho_D}{(N-1)M_N}. \tag{A.6}$$

A.5 Real exchange rates

Let $P_i = P_{Ti}^{\alpha_T} P_{Ni}^{1-\alpha_T}$ be i's consumer price index in its own currency. With P_{Ni} fixed and $d\log P_{Ti} = (1-\alpha_D)\,d\log P_{Mi}$ at the symmetric benchmark, $d\log P_i = m\,d\log P_{Mi}$, and from $q_{Aj} = e_{Aj}P_j/P_A$,

$$d\log q_{Aj} = d\nu_j + m\,(d\log P_{Mj} - d\log P_{MA}).$$

Let A impose t and its partners impose none. With equal import weights, $d\log P_{MA} = d\nu_A^E + \bar{t}$; partner j's import prices move by $d\nu_k - d\nu_j$ for $k \notin \{A, j\}$ and by $-d\nu_j$ for A's good, so $d\log P_{Mj} = d\nu_A^E - \frac{N}{N-1}d\nu_j$. Hence

$$d\log q_{Aj} = \Big(1 - \frac{mN}{N-1}\Big)d\nu_j - m\bar{t}, \qquad d\log q_{Aj} - d\log q_A^E = \Big(1 - \frac{mN}{N-1}\Big)\lambda_{\text{rel}}\widetilde{t}_j, \tag{A.7}$$

with $d\log q_A^E \equiv \frac{1}{N-1}\sum_{j\neq A} d\log q_{Aj}$. Averaging the first expression and using Eq. (3.7) gives Eq. (C.3). For a unilateral tariff, $\bar{t} = d\tau/(N-1)$, and averaging,

$$\frac{d\log q_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right)\frac{\rho_D}{(N-1)M_N} - \frac{m}{N-1}. \tag{A.8}$$

A.6 Three-country illustration

At $N = 3$ the single bystander c is also the composite C of Proposition 3.4. With $\nu = (\nu_B, \nu_c)$, Eqs. (A.2) and (A.3) give the unilateral-tariff system

$$J = -\frac{mM_3}{2}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_D \\ \rho_D - \rho \end{pmatrix}, \qquad M_3 = \frac{3\alpha_D(\eta-1) + \rho + 1}{2}. \tag{A.9}$$

With $M_3 > 0$,

$$\frac{d\log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_D}{6M_3} < 0, \qquad \frac{d\log e_{Ac}}{d\tau} = \frac{\rho - 3\rho_D}{6M_3}, \qquad \frac{d\log e_{Bc}}{d\tau} = \frac{\rho}{3M_3} > 0. \tag{A.10}$$

Since $(-J)^{-1} = \frac{2}{3mM_3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ is entrywise positive under MML, Eq. (2.4) gives $d\nu_c/d\tau = \frac{2}{3mM_3}(F_B + 2F_c)$, with $F_c = m(\rho - \rho_D)/4$ and $F_B = -m(\rho + \rho_D)/4 < 0$. For $\rho_D < \rho < 3\rho_D$, $F_c > 0$ but $F_B + 2F_c = m(\rho - 3\rho_D)/4 < 0$: c's trade balance improves at the initial exchange rates, yet A appreciates against c (Fig. 2a).

B Proofs

Notation and the sign convention are those of the main text; $m \equiv \alpha_T(1-\alpha_D) > 0$ at the symmetric benchmark. Unless stated otherwise, derivatives are first-order and evaluated at a balanced free-trade baseline, using the linear system of Appendix A. Appendix C.2 gives the real-exchange-rate extension and its proof.

B.1 Multilateral adjustment and symmetric partners

Proof of Lemma 2.3. Free trade is not required. By homogeneity and Euler's theorem, $\sum_{l\in\mathcal{I}} \widetilde{J}_{il} = \mathrm{TB}_i = 0$ at the balanced baseline. Since $\nu_l = \chi_l - \chi_A$ with χ_A fixed, J is $\widetilde{J}$ with row and column A deleted. Its off-diagonal entries are nonnegative, and for each $i \in \mathcal{I}_{-A}$

$$|J_{ii}| - \sum_{l\in\mathcal{I}_{-A}, l\neq i} J_{il} = \widetilde{J}_{iA} \geq 0,$$

strictly in at least one row. The diagonal of $-J$ is strictly positive: for $N = 2$, $|J_{BB}| = \widetilde{J}_{BA} > 0$; for $N \geq 3$, $J_{ii} = 0$ would force the nonnegative off-diagonal entries of row i to vanish, contradicting irreducibility. Write $-J = D - K$ with $D = \mathrm{diag}(-J) > 0$ and $K \geq 0$. The matrix $D^{-1}K$ is nonnegative and irreducible, with every row sum at most one and at least one strictly below one. The spectral radius of an irreducible nonnegative matrix is at most its largest row sum, with equality only if all row sums are equal (Perron–Frobenius), so the spectral radius of $D^{-1}K$ is below one and

$$(-J)^{-1} = (I - D^{-1}K)^{-1}D^{-1} = \sum_{k\geq 0}(D^{-1}K)^k D^{-1} \geq 0.$$

Hence MML holds. Because $D^{-1}K$ is irreducible, the series is entrywise positive, and $\det(-J) = \det D \cdot \det(I - D^{-1}K) > 0$. □

Proof of Proposition 2.4. No demand system, free-trade baseline or stability condition is used. If $PMP' = M$ for every $(N-1)\times(N-1)$ permutation matrix P, conjugating by permutations shows that all diagonal entries are equal and all off-diagonal entries are equal, so $M = aI + b\mathbf{1}\mathbf{1}'$, with eigenvalue $a + (N-1)b$ on $\mathrm{span}\{\mathbf{1}\}$ and a on the zero-sum subspace. For $-J = a_J I + b_J \mathbf{1}\mathbf{1}'$ and $G^{(A)} = a_G I + b_G \mathbf{1}\mathbf{1}'$ this gives $j_S = a_J + (N-1)b_J$, $j_D = a_J$, $g_S = a_G + (N-1)b_G$ and $g_D = a_G$, and for $j_S, j_D \neq 0$

$$(-J)^{-1} = \frac{1}{j_D} I + \frac{1}{N-1}\Big(\frac{1}{j_S} - \frac{1}{j_D}\Big)\mathbf{1}\mathbf{1}'.$$

Thus $(-J)^{-1}$ and $G^{(A)}$ share the two eigenspaces, and applying them to $t = \bar{t}\mathbf{1} + \widetilde{t}$ gives Eq. (2.6). Partner symmetry makes the baseline flows $f_{Aj} + f_{jA}$ equal across j, so $w_{Aj} = 1/(N-1)$; since $\widetilde{t}$ sums to zero, $d\nu_A^E = \lambda_{\mathrm{avg}}\bar{t}$, and the other two parts of Eq. (2.7) follow directly.

For the MML characterization, the off-diagonal entry of J is $-b_J = (j_D - j_S)/(N-1)$, so the sign pattern holds if and only if $j_D \geq j_S$. If $j_D \geq j_S > 0$, $-J$ is nonsingular and every entry of $(-J)^{-1}$ above is nonnegative, so MML holds. Conversely, under MML the sign pattern gives $j_D \geq j_S$, nonsingularity gives $j_S \neq 0$, and $(-J)^{-1}\mathbf{1} = \mathbf{1}/j_S \geq 0$ gives $j_S > 0$. □

B.2 The symmetric nested-CES model

The baseline is the symmetric benchmark of Section 3.1 with $N \geq 3$; for $N = 2$, Theorem 3.1 reduces to Eq. (3.4).

Proof of Theorem 3.1. The benchmark is invariant to relabeling A's partners, so Proposition 2.4 applies, and the eigenvalues Eq. (3.6) are those of Lemma A.1 and Eq. (A.4). Since $m > 0$, $j_S = mM_N/(N-1) > 0$ if and only if $M_N > 0$, and then $j_D = Nj_S > j_S$. By Proposition 2.4, MML is therefore equivalent to $M_N > 0$; if $M_N \leq 0$, then $j_S \leq 0$ and MML fails. If $\eta \geq 1$, the bracket in Eq. (3.5) is 1 plus a nonnegative term plus $(N-2)\rho > 0$, so $M_N > 0$. Given $M_N > 0$, dividing Eq. (A.4) by the eigenvalues of $-J$ gives $\lambda_{\text{avg}} = -\rho_D/M_N$ and $\lambda_{\text{rel}} = -\rho/(NM_N)$, and Eq. (3.7) is Eq. (2.6). □

Proof of Proposition 3.2. Equation (3.8) follows from Eq. (3.7) (Appendix A.4). Under MML $M_N > 0$, and $\rho_D > 0$ since $\alpha_D \eta > 0$; hence $d\nu_A^E/d\tau < 0$ and, since $N\rho_D + (N-2)\rho > 0$, $d\nu_B/d\tau < 0$. The denominator of $d\nu_c/d\tau$ is positive, so $d\nu_c/d\tau > 0$ if and only if $\rho > N\rho_D$, with $d\nu_c = 0$ at $\rho = N\rho_D$. For (iv), Eq. (A.6) gives each untargeted partner k times the unilateral response, so the threshold is again $\rho > N\rho_D$. For a general tariff vector with $\bar{t} > 0$, Eq. (A.5) gives the stated condition; when $t_j \geq 0$ its left side is nonnegative, so it can hold only if $1 - N\rho_D/\rho > 0$, that is, $\rho > N\rho_D$, and it then implies $t_j < \bar{t}$. □

Proof of Corollary 3.3. Equation (3.9) is an identity (Appendix A.4); under $\alpha_D \geq 1/N$ both terms on its right are nonnegative, so $N\rho_D \geq \eta$, and since a bystander depreciation requires $\rho > N\rho_D$ (Proposition 3.2), it requires $\rho > \eta$. With $\rho = \eta = \sigma$, Eq. (3.8) gives $d\nu_c/d\tau = (\sigma - N\rho_D)/[N(N-1)M_N]$ with $\sigma - N\rho_D = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)] \leq 0$; MML is used only to give $M_N > 0$, so $d\nu_c/d\tau \leq 0$. Since $\sigma > 0$ and $\alpha_D < 1$, the bracket, a sum of two nonnegative terms, vanishes if and only if $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$, $M_N = \sigma > 0$, so MML holds for every $\sigma > 0$, and $d\nu_c/d\tau = -(1-\alpha_T)/(N\sigma)$, which is negative and tends to zero as $\sigma \to \infty$ if $\alpha_T < 1$ and is identically zero if $\alpha_T = 1$. □

Proposition 3.4 applies to any marginal tariff change that is invariant to permutations of the $N-2$ bystanders and leaves tariffs among bystanders unchanged: tariffs by A or B common across bystanders, tariffs by all bystanders on A or on B common across bystanders, and tariffs between A and B. The mapping sets $d\tau_{AC} = d\tau_{Ac}$, $d\tau_{CA} = d\tau_{cA}$, $d\tau_{BC} = d\tau_{Bc}$, $d\tau_{CB} = d\tau_{cB}$ (the common values) and retains $d\tau_{AB}, d\tau_{BA}$. In the three-region economy, C has income $y_C = N-2$, home share h_C with $(N-1)h_C = N - 3 + 2\alpha_D$, weight $\frac{1}{2}$ on each of A and B in its import nest, tradable share α_T, cross-origin elasticity ρ and home–import elasticity η_C; A and B place import weight $\frac{1}{N-1}$ on each other and $\frac{N-2}{N-1}$ on C.

Proof of Proposition 3.4. The argument compares the two linearized systems at the benchmark; it does not claim that the nonlinear economies coincide at finite tariffs. MML is sufficient for $M_N \neq 0$ but is not otherwise used.

Common bystander response. By Lemma A.1, $-J = j_S(NI - \mathbf{1}\mathbf{1}')$ with eigenvalues j_S and Nj_S, and $j_S = mM_N/(N-1) \neq 0$, so Eq. (2.2) has a unique solution. Baseline and shock are invariant to permutations of the bystanders, hence so is the solution: $d\nu_c = d\nu_C$ for all c. The equilibrium is then given by $d\mathrm{TB}_B = 0$ and $\sum_c d\mathrm{TB}_c = 0$ in $(d\nu_B, d\nu_C)$.

Flows and A's and B's demand. In the N-country economy $\sum_c f_{Ac} = \sum_c f_{cA} = m\frac{N-2}{N-1}$; in the composite $f_{AC} = m\frac{N-2}{N-1}$ and, since $1-h_C = 2(1-\alpha_D)/(N-1)$, $f_{CA} = \alpha_T(1-h_C)\frac{1}{2}(N-2) = m\frac{N-2}{N-1}$; likewise for B, and $f_{AB} = f_{BA} = m/(N-1)$ in both, so $s_{CA} = s_{cA} = m/(N-1)$. All bystander goods face the common price change $d\nu_C + d\tau_{Ac}$ in A, so A's import price index, its common share change $d\log s_{Ac}$ and its income response coincide with the composite's, and $\sum_c df_{Ac} = df_{AC}$; the same holds for B.

Intra-bloc trade cancels. Summing Eq. (3.3) over bystanders, the terms $\sum_c \sum_{c' \neq c}(f_{c'c} - f_{cc'})$ cancel identically, so $\sum_c \mathrm{TB}_c$ has the form of $\mathrm{TB}_C = f_{AC} + f_{BC} - f_{CA} - f_{CB}$, and TB_B has the form of the composite's. It remains to show $\sum_c df_{cA} = df_{CA}$ and $\sum_c df_{cB} = df_{CB}$.

Identification of y_C, h_C and η_C. Since $d\log f_{cA} = d\log s_{cA} + d\log \tilde{I}_c - d\tau_{cA}$ is common across c, $\sum_c df_{cA} = f_{CA}\, d\log f_{cA}$, which fixes $y_C = N-2$. Incomes match because $s_{CA} = s_{cA}$ and $s_{CB} = s_{cB}$, and $d\tau_{cA} = d\tau_{CA}$. For shares, measure prices in bystander currency: because bystander currencies move together and tariffs among bystanders do not change, other bystanders' goods have $dp = 0$. Write $S \equiv dp_{cA} + dp_{cB}$ and $D \equiv dp_{cA} - dp_{cB}$. A bystander's import index moves by $S/(N-1)$, the composite's by $S/2$, so

$$
\begin{aligned}
d\log s_{cA} &= (1-\rho)\frac{D}{2} + \left[(1-\rho)\frac{N-3}{2(N-1)} + (1-\eta)\frac{\alpha_D}{N-1}\right]S, \\
d\log s_{CA} &= (1-\rho)\frac{D}{2} + (1-\eta_C)h_C\frac{S}{2}.
\end{aligned}
$$

The D terms coincide for all (S, D); the S terms coincide if and only if $(1-\eta_C)(N-1)h_C = (1-\rho)(N-3) + 2\alpha_D(1-\eta)$, which, with $(N-1)h_C = N-3+2\alpha_D$, is Eq. (3.11). The argument for B's good replaces D by $-D$, and the residual share matches because shares sum to one.

Restricted Jacobian and solution. Hence the two equilibrium conditions, including tariff terms, are identical to the composite's. Imposing $d\nu_c = d\nu_C$ on row B and on the sum of the bystander rows of $-J$,

$$
-J = j_S \begin{pmatrix} N-1 & -(N-2) \\ -(N-2) & 2(N-2) \end{pmatrix}, \qquad \det(-J) = j_S^2 N(N-2), \tag{B.1}
$$

which for $N \geq 3$ is nonsingular if and only if $j_S \neq 0$, whatever its sign. With disturbances

F_B, F_c and $F_C = (N-2)F_c$,

$$d\nu_C = \frac{F_B + (N-1)F_c}{Nj_S}, \qquad d\nu_B = \frac{2F_B + (N-2)F_c}{Nj_S}, \tag{B.2}$$

the unique solution of both systems. The composite's trade weights $\frac{1}{N-1}$ and $\frac{N-2}{N-1}$ reproduce A's NEER at symmetric weights. For $N = 3$, $h_C = \alpha_D$ and $\eta_C = \eta$, so C is the single bystander. □

For the unilateral tariff, $F_B + (N-1)F_c = m(\rho - N\rho_D)/(N-1)^2$ (Appendix A.4), so Eq. (B.2) reproduces the threshold of Proposition 3.2. If instead the composite keeps the mapped y_C, h_C, flows, weights and ρ but replaces η_C by η, it no longer reproduces the N-country economy for $N \geq 4$, and its bystander threshold, computed from the linear system, falls slightly with N: at the US-like calibration ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$), from 3.96 at $N = 3$ to 3.71 at $N = 8$, while $N\rho_D$ rises from 3.96 to 10.56. The steep N-dependence of $N\rho_D$ therefore comes from the bloc's internal trade, governed by ρ, rather than from its size.

B.3 Retaliation, welfare, and asymmetry

Proof of Proposition 4.1. Under MML $M_N > 0$ (Theorem 3.1), and the response is linear in tariffs, so responses to simultaneous tariffs superpose. (i) Shock 1, $d\tau_{AB} = d\tau$, is Eq. (3.8). Shock 2, $d\tau_{BA} = d\tau$, is the same experiment with A and B exchanged, so $d\log e_{BA} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}d\tau$ and $d\log e_{Bc} = \frac{\rho - N\rho_D}{N(N-1)M_N}d\tau$; with $e_{AB} = 1/e_{BA}$ and $e_{Ac} = e_{AB}e_{Bc}$,

$$d\nu_B^{(2)} = \frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\,d\tau, \qquad d\nu_c^{(2)} = \frac{\rho}{NM_N}\,d\tau.$$

Adding shock 1, $d\nu_B = 0$ and

$$\frac{d\nu_c}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N} + \frac{\rho}{NM_N} = \frac{\rho - \rho_D}{(N-1)M_N}$$

for every bystander, so cross rates among bystanders are unchanged and $d\log e_{Bc} = d\nu_c - d\nu_B = d\nu_c$. With equal weights, $d\nu_A^E = \frac{N-2}{N-1}d\nu_c$, and B's NEER, $[0 + (N-2)d\nu_c]/(N-1)$, is the same. Since $M_N > 0$ and $N \geq 3$, every response has the sign of $\rho - \rho_D$.

(ii) A's uniform tariff contributes $\lambda_{\text{avg}}d\tau = -\frac{\rho_D}{M_N}d\tau$ to every $d\nu_j$. Partner j's tariff on A is shock 2 with j in the role of B: it contributes $\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}d\tau$ to $d\nu_j$ and $\frac{\rho}{NM_N}d\tau$ to each of

the other $N-2$ partners. Summing,

$$\frac{d\nu_j}{d\tau} = -\frac{\rho_D}{M_N} + \frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} + \frac{(N-2)\rho}{NM_N} = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N},$$

common to all j, so cross rates among partners are unchanged and $d\nu_A^E$ equals this response, with the sign of $\rho - \rho_D$. □

Proof of Lemma 5.1. Indirect utility is $V_A = \kappa\, I_A/(P_{TA}^{\alpha_T} P_{NA}^{1-\alpha_T})$, so the welfare change in units of income is $d\log I_A - \alpha_T\, d\log P_{TA} - (1-\alpha_T)\, d\log P_{NA}$. By Shephard's lemma, $\alpha_T\, d\log P_{TA} = \sum_j s_{Aj}\, d\log p_{Aj}$. Under the producer-price normalization P_{NA} and the domestic tradable's price are fixed in A's currency, and at free trade $d\log p_{Aj} = d\nu_j + d\tau_{Aj}$. Income is $I_A = w_A L_A/(1-\sum_j \theta_{Aj} s_{Aj})$ with $\theta_{Aj} = \tau_{Aj}/(1+\tau_{Aj})$; at free trade $\theta_{Aj} = 0$ and $d\theta_{Aj} = d\tau_{Aj}$, so share changes drop out and $d\log I_A = \sum_j s_{Aj}\, d\tau_{Aj}$. Combining,

$$\frac{dU_A}{U_A' I_A} = \sum_j s_{Aj}\, d\tau_{Aj} - \sum_j s_{Aj}(d\nu_j + d\tau_{Aj}) = -m_A \sum_{j\neq A} b_{Aj}\, d\nu_j,$$

with $m_A = \sum_j s_{Aj}$ and $b_{Aj} = s_{Aj}/m_A$. The tariff terms cancel because $\tau_{Aj} = 0$ at the baseline; the cancellation is first order only. Other countries' tariffs enter A's prices and income only through $d\nu$, so the formula holds for any tariff vector. □

Proof of Proposition 5.2. Use Eq. (A.1) with country-specific η_i, ρ_i and α_T^i, so $s_{Aj} = \alpha_T^A(1-h_A)b_{Aj}$ and $m_A = \alpha_T^A(1-h_A)$. At $d\nu = 0$, no country other than A faces a price change, so every flow f_{ij} with $i \neq A$ is unchanged and $F_j = f_{Aj}(d\log s_{Aj} + d\log \tilde{I}_A - dt)$. In A, $dp_{Aj} = dt$ for every j, so $d\log P_{MA} = dt$, the cross-origin term vanishes, $d\log s_{Aj} = (1-\eta_A)h_A\, dt$ and $d\log \tilde{I}_A = m_A\, dt$. Hence

$$F_j = f_{Aj}\Big[(1-\eta_A)h_A + \alpha_T^A(1-h_A) - 1\Big]dt = -f_{Aj}\,\rho_D^A\, dt,$$

which is Eq. (5.4); $\rho_D^A > 0$ since $\eta_A > 0$, $0 < h_A \leq 1$ and $\alpha_T^A \leq 1$, so $F_j \leq 0$, strictly if $f_{Aj} > 0$. Under MML, Eq. (2.4) gives $d\nu_i \leq 0$ for every i. If A imports from every partner, $F < 0$, and each row of the nonsingular nonnegative $(-J)^{-1}$ has a positive entry, so $d\nu_i < 0$ for every i; if J is irreducible, $(-J)^{-1} > 0$ entrywise (Lemma 2.3), so $d\nu_i < 0$ for every i whenever $F \neq 0$. By Lemma 5.1, the welfare change $-m_A \sum_j b_{Aj} d\nu_j$ is nonnegative, and strictly positive in either case since the b_{Aj} sum to one; the aggregate terms of trade improve for the same reason. By linearity this welfare change is $\sum_j g_j\, dt$, so $\sum_j g_j > 0$. □

Proof of Proposition 5.4. At any ρ at which MML holds, $-J$ is a nonsingular M-matrix, so $\det A(\rho) > 0$ and $R_{A|B}$ has the sign of $Q_{A|B}(\rho)$. (a) At the symmetric benchmark, Eq. (A.9)

gives $\det A = 3(mM_3/2)^2$ and $R_{A|B} = -\rho_D/(2M_3)$, so $Q = -\frac{3}{8}m^2\rho_D M_3$, affine in ρ ($q_2 = 0$) and negative wherever MML holds. (b) If $q_2 > 0$ and $q_0 < 0$, the product of the roots q_0/q_2 is negative, so there is exactly one positive root ρ^*, with $Q < 0$ on $(0, \rho^*)$ and $Q > 0$ beyond. (c) If $q_2 < 0$, $q_1 > 0$, $q_0 < 0$ and the discriminant is positive, the roots are real with positive sum $-q_1/q_2$ and positive product q_0/q_2, and the downward parabola is positive only between them. The cases are not exhaustive: other sign patterns, such as $q_0 > 0$, or $q_2 < 0$ with no positive root, are possible; in the latter case with $q_0 < 0$, $Q_{A|B} < 0$ and the condition holds at every ρ at which MML holds. □

B.4 Verification

The analytical Jacobians and tariff disturbances were compared with centered finite differences of the nonlinear model, at symmetric and asymmetric baselines. The closed forms in this appendix and in Appendix A were checked against direct solutions of the linear system, and every reported threshold was checked on both sides, with MML verified at each point.

C Departures from the Symmetric Free-Trade Benchmark

This appendix extends the local results of Sections 2.4, 3 and 4 to common and finite tariffs, real exchange rates and asymmetric primitives, and documents the U.S.–China–rest-of-world mapping and the 2025 data. Equation (2.3) and the MML condition apply at any balanced baseline. Values not derived analytically are numerical comparative statics computed from the exact model, with the baseline solved from primitives; they are not global theorems. MML holds at every reported root and point. Numerical values use the US-like parameters $(\alpha_D, \alpha_T, \eta) = (0.8, 0.4, 1.5)$, $\rho_D = 1.32$.

C.1 Common and finite tariffs

Suppose A levies a common tariff τ_0 on all $N-1$ partners, which impose no tariffs of their own. With symmetric partner primitives this baseline is invariant to every relabeling of A's partners, so partner symmetry holds and Proposition 2.4 applies. Tariff changes are measured as $t_j = d\log(1+\tau_{Aj})$. Let $\theta \equiv \tau_0/(1+\tau_0)$, μ_A and μ_P be the import shares of A's and of a partner's tradable expenditure, ω the share of A's good in a partner's import bundle, $\phi \equiv \alpha_T\mu_A/(1-\theta\alpha_T\mu_A)$ the response of A's income to its import expenditure, $X \equiv f_{Aj} = f_{jA}$ the balanced bilateral flow between A and each partner, and $Y \equiv f_{jk}$ the flow between two

partners. Linearizing at this baseline gives

$$g_S = -X\rho_D(\tau_0), \qquad g_D = -X\rho, \qquad j_S = XM_S, \qquad j_D = \Big(X + (N-1)Y\Big)M_D, \quad \text{(C.1)}$$

where

$$\begin{aligned}
\rho_D(\tau_0) &= 1 - \phi(1-\theta) + (\eta-1)(1-\mu_A)(1+\phi\theta),\\
M_S &= 1 + (\eta-1)(1-\mu_A)(1+\phi\theta) + (\rho-1)(1-\omega) + (\eta-1)(1-\mu_P)\,\omega,\\
M_D &= 1 + (\rho-1)\frac{N-3+\omega}{N-2} + (\eta-1)(1-\mu_P)\frac{N-1-\omega}{N-2}.
\end{aligned}$$

At free trade $\theta = 0$, $\phi = m$, $\mu_A = \mu_P = 1 - \alpha_D$ and $\omega = 1/(N-1)$, so $\rho_D(0) = \rho_D$ and $M_S = M_D = M_N$. Tariff disturbances and common exchange-rate movements act only on partners' trade with A, whereas a zero-sum movement v also moves partner j's rate against the other partners, on average by $(N-1)/(N-2)$ times v_j; hence the bases X and $X + (N-1)Y$. With $x \equiv X/(X + (N-1)Y)$, $\lambda_{\text{avg}} = -\rho_D(\tau_0)/M_S$ and $\lambda_{\text{rel}} = -x\rho/M_D$. For a further tariff on B, $t = dt\, e_B$, the bystander response is $d\nu_C = (\lambda_{\text{avg}} - \lambda_{\text{rel}})\, dt/(N-1)$, so A depreciates against C if and only if

$$\rho > \rho^*(\tau_0) \equiv \frac{\rho_D(\tau_0)}{x}\,\frac{M_D}{M_S}. \quad \text{(C.2)}$$

At free trade $x = 1/N$, so the factor N in the bystander threshold is $1/x$ at the free-trade baseline. Because ω, μ_P, x and M_S depend on baseline shares that depend on ρ, Eq. (C.2) is implicit in ρ and is solved numerically with the tariff-ridden baseline re-solved at each ρ. By linearity in t, a marginal preferential cut for B, $t = -dt\, e_B$, worsens C's terms of trade under the same condition.

In the reported exercises a common external tariff raises the threshold. At $N = 3$, $\rho^*(\tau_0)$ is about 10 percent above its free-trade value $N\rho_D$ at $\tau_0 = 0.25$ and about 36 percent above it at $\tau_0 = 1$. The main reason is that x falls: the common appreciation of A's currency lowers A's share of each partner's imports, so relative treatment disturbs a smaller part of the trade on which relative exchange-rate movements act. The local threshold applies to marginal cuts; in the exact model, removing the tariff on B entirely requires a higher ρ than a marginal cut at the same baseline.

A concentrated finite tariff works in the opposite direction. At $N = 3$, starting from the symmetric free-trade benchmark and re-solving the equilibrium at each tariff, the unilateral bystander threshold falls with the size of the tariff in the computed cases: at the 2025 peak rate $\tau = 1.45$ it is 2.70, against $3\rho_D = 3.96$. A larger tariff shrinks B's share of A's imports,

so a further increase adds less average protection while continuing to divert expenditure toward C.

C.2 Real exchange rates

Let P_i be i's consumer price index in its own currency and $q_{Aj} \equiv e_{Aj}P_j/P_A$ the real exchange rate, so that a rise is a real depreciation of A. At the symmetric benchmark the real effective exchange rate is $d\log q_A^E \equiv \frac{1}{N-1}\sum_{j\neq A} d\log q_{Aj}$. With partners imposing no tariffs, to first order (Appendix A.5)

$$d\log q_{Aj} = \Big(1 - \tfrac{mN}{N-1}\Big)d\nu_j - m\bar{t}, \qquad d\log q_A^E = \Big[\Big(1 - \tfrac{mN}{N-1}\Big)\lambda_{\text{avg}} - m\Big]\bar{t}. \tag{C.3}$$

The term $-m\bar{t}$ is the direct effect of the tariff on A's consumer price index, which offsets part of a nominal depreciation.

Proposition C.1 (Real exchange rates)**.** *Consider a marginal unilateral tariff $d\tau > 0$ by A on B at the symmetric benchmark, $N \geq 3$, $m > 0$, partners imposing no tariffs, and define*

$$H_N(\rho) \equiv \Big(1 - \frac{mN}{N-1}\Big)(\rho - N\rho_D) - mN\, M_N(\rho).$$

For every untariffed bystander c, if $M_N(\rho) \neq 0$, $d\log q_{Ac}/d\tau = H_N(\rho)/[N(N-1)M_N(\rho)]$, so wherever MML holds the real bilateral response has the sign of $H_N(\rho)$. Since ρ_D does not depend on ρ, $dH_N/d\rho = 1 - mN$.

(i) If $m < 1/N$, H_N has the unique root

$$\rho_q^*(N) = \frac{[(N-1) - mN]N\rho_D + mN[N\alpha_D(\eta-1)+1]}{(N-1)(1-mN)}.$$

Where MML holds, A appreciates in real terms against c for $\rho < \rho_q^*(N)$ and depreciates for $\rho > \rho_q^*(N)$; if MML holds at $\rho_q^*(N)$, then $\rho_q^*(N) > N\rho_D$.

(ii) If $m = 1/N$, H_N does not depend on ρ: $H_N = -[1+N\alpha_D(\eta-1)+N(N-2)\rho_D]/(N-1)$.

(iii) If $m > 1/N$, H_N is decreasing in ρ, so any crossing has the reverse orientation.

(iv) If $\eta \geq 1$, so that MML holds for every $\rho > 0$, and $m \geq 1/N$, then $H_N(\rho) < 0$ for every $\rho > 0$: no positive ρ produces a real depreciation against an untariffed bystander.

(v) Under MML, if $m \leq (N-1)/N$, A's REER appreciates. Under weak home bias this condition holds automatically, since $m = \alpha_T(1-\alpha_D) \leq (N-1)/N$.

At the US-like calibration and $N = 3$, $m = 0.08 < 1/3$ and $\rho_q^* = 4.93$, against the nominal threshold $3\rho_D = 3.96$.

Proof of Proposition C.1. MML is used only for the sign statements in (i), (iii) and (v) and for the comparison in (i); in (iv) it follows from $\eta \geq 1$. By Eq. (A.7) with $\bar{t} = d\tau/(N-1)$, $d\log q_{Ac}/d\tau = (1 - \frac{mN}{N-1})\, d\nu_c/d\tau - \frac{m}{N-1}$. Substituting $d\nu_c/d\tau$ from Eq. (3.8) and writing $\frac{m}{N-1} = \frac{mN\,M_N}{N(N-1)M_N}$ gives the stated response. Under MML $M_N > 0$, so the response has the sign of H_N. Since M_N in Eq. (3.5) has slope $(N-2)/(N-1)$, H_N is affine in ρ with slope $1 - \frac{mN}{N-1} - \frac{mN(N-2)}{N-1} = 1 - mN$.

(i) For $m < 1/N$ the slope is positive, so H_N has a unique root, is negative below it and positive above it. Multiplying $H_N = 0$ by $N-1$ gives $[(N-1) - mN](\rho - N\rho_D) = mN[1 + N\alpha_D(\eta-1) + (N-2)\rho]$; the coefficient on ρ is $(N-1) - mN - mN(N-2) = (N-1)(1-mN)$, which yields $\rho_q^*(N)$. At the root $[(N-1) - mN](\rho_q^* - N\rho_D) = mN(N-1)M_N(\rho_q^*)$. For $m < 1/N$ the coefficient $(N-1) - mN$ exceeds $N - 2 \geq 1$, and if MML holds at ρ_q^* the right-hand side is strictly positive because $m > 0$ and $M_N(\rho_q^*) > 0$; hence $\rho_q^*(N) > N\rho_D$, strictly.

(ii) For $m = 1/N$ the slope is zero, and $H_N = \frac{N-2}{N-1}(\rho - N\rho_D) - \frac{1 + N\alpha_D(\eta-1) + (N-2)\rho}{N-1}$, which reduces to the stated constant.

(iii) For $m > 1/N$ the slope is negative, so H_N has at most one root and is positive below it and negative above it.

(iv) If $\eta \geq 1$, every term of M_N in Eq. (3.5) is nonnegative and the first is positive, so MML holds for every $\rho > 0$. Since $(N-1)M_N(0) = 1 + N\alpha_D(\eta-1)$ and $\rho_D = 1 + \alpha_D(\eta-1) - m$,

$$(N-1)H_N(0) = -[(N-1) - mN]N\rho_D - mN[1 + N\alpha_D(\eta-1)] = -N\Big[(N-1)(\rho_D - m) + Nm^2\Big].$$

With $\eta \geq 1$, $\rho_D - m \geq 1 - 2m$, so

$$(N-1)(\rho_D - m) + Nm^2 \geq Nm^2 - 2(N-1)m + (N-1) = N\left(m - \frac{N-1}{N}\right)^2 + \frac{N-1}{N} > 0.$$

Hence $H_N(0) < 0$. For $m \geq 1/N$, $dH_N/d\rho \leq 0$, so $H_N(\rho) \leq H_N(0) < 0$ and $d\log q_{Ac}/d\tau < 0$ for every $\rho > 0$.

(v) Under MML $M_N > 0$, and $\rho_D > 0$. In Eq. (A.8), the first term is nonpositive if and only if $1 - mN/(N-1) \geq 0$, that is, $m \leq (N-1)/N$, and the second, $-m/(N-1)$, is strictly negative because $m > 0$. Hence the REER strictly appreciates whenever $m \leq (N-1)/N$, including at $m = (N-1)/N$. Under weak home bias, $1 - \alpha_D \leq (N-1)/N$ and $\alpha_T \leq 1$, so $m = \alpha_T(1 - \alpha_D) \leq (N-1)/N$. □

C.3 Asymmetric primitives and effective rates

At a general balanced baseline the coefficients of Eq. (A.1) are the baseline shares b_{ij}, the home shares h_i and the incomes y_i. The elasticity ρ enters the share responses only through $(1-\rho)$, so every entry of J and F is affine in ρ, each entry of $\mathrm{adj}(-J)$ is a polynomial of degree at most $N-2$ in ρ, and each numerator $[\mathrm{adj}(-J)F]_i$ a polynomial of degree at most $N-1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator. At a general asymmetric baseline the $N=3$ numerator is already quadratic in ρ:

$$\operatorname{sign}\frac{d\log e_{AC}}{d\tau} = \operatorname{sign}\left(c_2\rho^2+c_1\rho+c_0\right), \quad c_2 = \alpha_T^2\, b_{AB}b_{AC}b_{CA}b_{CB}(1-h_A)(1-h_C)y_Ay_C, \tag{C.4}$$

where c_1 and c_0 depend on the full pattern of shares and sizes. Provided MML holds, $c_2 > 0$ whenever A and C each import from both foreign origins, so at given shares the response is positive beyond the largest positive root, which equals $3\rho_D$ at the symmetric point.

Once primitives rather than shares are held fixed, the shares themselves depend on ρ: at an asymmetric baseline relative prices differ across origins, and a higher ρ reallocates expenditure toward the cheapest origin. A large target, whose currency depreciates to balance its trade, can then crowd C out of A's imports. With all other primitives symmetric and the baseline re-solved at each $\rho \in [1.02, 1000]$, e_{AC} reverses only for $5.99 < \rho < 53.8$ when $L_B = 2.2$ and nowhere when $L_B = 2.6$. Away from symmetry there may therefore be no bilateral threshold, one or several. Near the symmetric benchmark, the bilateral threshold rises with A's origin weight on the target (a larger average-protection shock), with the target's size (which raises its share in A's imports) and with home bias in A alone (stronger domestic diversion), and is nearly insensitive to the bystander's size.

The trade-weighted NEER is $d\nu_A^E = \sum_{j\neq A} w_{Aj}\, d\nu_j$. At the symmetric benchmark the zero-sum component averages to zero and under MML a marginal unilateral tariff appreciates the NEER. For $N=3$ with asymmetric weights, $d\nu_A^E = w_{AB}\, d\nu_B + w_{AC}\, d\nu_C$, so the NEER can depreciate when the bystander reversal occurs and the bystander carries most of A's trade. The symmetric ordering $d\nu_B < d\nu_C$ can also fail: with $L = (5.12, 4.26, 2.28)$, $\beta_{AB} = 0.86$, $\beta_{BA} = 0.65$, $\beta_{CA} = 0.19$, $\alpha_D = 0.89$, $\alpha_T = 0.6$, $\eta = 3.74$ and $\rho = 1.20$, $d\nu_B - d\nu_C = 0.08\, d\tau$.

The three panels of Fig. 5 use the US-like parameters. In (a) and (b) the free-trade baseline is solved from primitives at $\bar\rho = 3\rho_D = 3.96$ and then held fixed; $\hat Q$ is $Q_{A|B}$ in Eq. (5.7) scaled by $|q_2|$ (by $|q_1|$ when $q_2 = 0$), which preserves its sign and roots, and MML holds at all three baselines for $\rho \in [0.01, 200]$. (a) Symmetric benchmark: $\hat Q = -(\rho + 2.2)$. (b) $L = (1, 0.22, 1)$, $\beta_{AB} = 0.2$, $\beta_{AC} = 0.8$, other origin weights symmetric ($b_{AB} = 0.094$): $\hat Q = \rho^2 - 2.6695\rho - 2.2568$, with root $\rho^* = 3.344$. (c) A network specified in

baseline shares, with $\alpha_T = 0.862$, $\eta = 3.007$, home shares $(0.201, 0.854, 0.360)$, $(b_{AB}, b_{AC}) = (0.525, 0.475)$, $(b_{BA}, b_{BC}) = (0.766, 0.234)$, $(b_{CA}, b_{CB}) = (0.407, 0.593)$ and balancing incomes $(1, 5.126, 0.867)$: $\hat{Q} = -\rho^2 + 31.53\rho - 198.04$, so the condition fails for $8.66 < \rho < 22.86$.

In the family containing example (b), with $L_A = L_C = 1$ and $\beta_{BA} = \beta_{CA} = 0.5$, and with baselines fixed at $\bar{\rho}$, $q_0 < 0$ throughout. A positive q_2, and hence case (b) of Proposition 5.4, arises when the target is small, has a low import weight, or both, with an approximate trade-off between the two; the root rises as the target becomes larger or more heavily weighted. For a large or heavily weighted target $q_2 < 0$ and $Q_{A|B}$ has no positive root, so the condition holds at every ρ at which MML holds.

The three thresholds are distinct. With $\beta_{AB} = 0.2$, $L_B = 0.223$ (so that $w_{AC} = 0.85$ at $\rho = 3\rho_D$), all other primitives symmetric and the baseline re-solved at each ρ, A depreciates against the bystander for $\rho > 2.59$, its aggregate terms of trade deteriorate ($R_{A|B} > 0$) for $\rho > 3.40$, and its NEER depreciates for $\rho > 3.97$. The bystander threshold lies below $3\rho_D$ because the target is small and lightly weighted. The NEER threshold is highest because the target's weight in A's trade, $w_{AB} = 0.15$, exceeds its weight in A's imports, $b_{AB} = 0.095$, so the favorable move against the target counts for more in the NEER. A larger bystander weight w_{AC}, obtained with a smaller target, lowers all three thresholds.

With $N \geq 4$ and bystanders that differ from one another, partition the reduced system into the target B and the set R of the $N-2$ bystanders. Correspondingly, J_{RR} is the block of J giving the bystanders' trade-balance responses to their own rates, J_{RB} the column giving their responses to $d\nu_B$, F_R their tariff disturbances per unit of $d\tau$, and $d\nu_R$ the vector of A's bilateral rates against the bloc. Let $q_j \equiv b_{Aj}/(1 - b_{AB})$, $j \in R$, be bystander j's share in A's imports from the bloc, so that $q'\mathbf{1} = 1$ and $q'd\nu_R/d\tau = \bar{R}^B_{-B}$. The bystanders' rows of the local system read $J_{RB}\, d\nu_B + J_{RR}\, d\nu_R + F_R\, d\tau = 0$. Under MML $-J$ is a nonsingular M-matrix, so its principal block $-J_{RR}$ is nonsingular as well. Solving for $d\nu_R$ and weighting by q gives the bloc's import-weighted response,

$$d\bar{\nu}_R \equiv q'd\nu_R = -q'J_{RR}^{-1}J_{RB}\, d\nu_B - q'J_{RR}^{-1}F_R\, d\tau. \tag{C.5}$$

C.4 Illustrative U.S.–China–rest-of-world mapping

This mapping places a unilateral US tariff on China in the three-country model, with A the United States, B China and C the rest of the world. It maps trade shares and country sizes; it is not a calibration of exchange-rate magnitudes.

All inputs are for 2024, in billions of current dollars. US goods imports are 3,266.4 in total and 440.3 from China, and US goods exports 2,061.7 in total and 143.3 to China (US Census Bureau). China's goods imports are 2,585.1 in total and 163.6 from the United States, and

its exports 3,577.2 in total and 524.7 to the United States (China's General Administration of Customs). This gives $b_{AB} = 0.135$ and $b_{BA} = 0.063$. The rest of the world's imports from the United States are US exports less US exports to China, and its imports from China are China's exports less China's exports to the United States, which gives $b_{CA} = 0.386$ and $b_{CB} = 0.614$. Country sizes are GDP at purchasing-power parity (IMF World Economic Outlook, October 2026), with the rest of the world equal to world GDP less that of the United States and China; relative to the United States they are $(1, 1.30, 4.45)$. The origin weights β_{ij} are chosen so that the balanced free-trade baseline, solved at $\bar{\rho} = 3\rho_D = 3.96$ and then held fixed as ρ varies, reproduces these six shares exactly. The remaining parameters are $\alpha_T = 0.4$ and $\eta = 1.5$, with either a common $\alpha_D = 0.8$ or, for the United States, $\alpha_D = 0.673$, the domestic share of US goods absorption in 2024 (BEA goods output plus imports less exports).

The bilateral bystander threshold, above which the dollar depreciates against the rest of the world, is 4.01 with the common home bias and 3.87 with the measured US home share. The aggregate terms-of-trade condition Eq. (5.5) fails only for ρ above 4.66 and 4.50, respectively. MML holds at both baselines for all $\rho \in [1, 20]$. The mapping imposes balanced trade on every country, whereas the observed US goods deficit in 2024 was about 1,205; it ignores the gaps between mirror trade statistics, uses model-implied rather than measured home shares for China and the rest of the world, covers goods only and treats the rest of the world as a single country.

C.5 Data notes for the 2025 episode

Exchange-rate data. Bilateral rates are Federal Reserve H.10 noon buying rates, expressed in dollars per unit of foreign currency (the CNY, CAD and MXN quotes are inverted). The broad index is the H.10 nominal broad dollar index (goods and services), which rises when the dollar appreciates; it is inverted. The index excluding China removes the renminbi from the daily log change of the broad index using China's 2025 H.10 weight (10.9 percent) and divides by one minus that weight, so the remaining weights are renormalized to sum to one. No daily dong series is available from the H.10 release or from the BIS; the VND row of Table 1 uses BIS monthly averages in the monthly columns and, in the daily columns (marked [a]), BIS end-of-month rates for the same months (end-January to end-March, end-March to end-April, end-July to end-December). Sources: FRED (DEXUSEU, DEXCHUS, DEXCAUS, DEXMXUS, DTWEXBGS), Federal Reserve H.10 trade weights and BIS (VND). "Monthly" compares monthly averages, January with March (Phase 1), March with April (Phase 2) and July with December (Phase 3), windows that do not coin-

cide with the phase dates; "daily" runs from January 31 to April 2, April 2 to May 12 and August 6 to December 31. Panel (a) of Fig. 6 plots monthly averages, at mid-month, relative to each series' January 2025 average; the August dong average is missing in the source.

Tariff data. Panel (b) of Fig. 6 plots statutory headline tariffs above the pre-2025 baseline by effective date; Section 232 product tariffs, exclusions and de minimis changes are omitted, and rates are not trade-weighted. Effective dates (2025), in percent: US on China 10 (February 4), 20 (March 4), 30 (April 5), 104 (April 9), 145 (April 10), 30 (May 14), 20 (November 10); China on US, across the board, 84 (April 10), 125 (April 12), 10 (May 14), with the product-specific measures of February 10 and March 10 not shown; US on EU 10 (April 5), 20 (April 9 only), 10 (April 10), 15 inclusive of MFN duties (August 7); US baseline on unlisted partners 10 (April 5); US on Canada 25 (March 4), non-USMCA goods only from March 7, 35 (August 1), with the rate on Mexico at 25 from March 4 to year-end (not shown); Canada on US 25 on covered goods (March 4), with coverage widened on March 13 and April 9 and narrowed to steel, aluminium and autos on September 1. US measures are from Federal Register executive orders and Canadian measures from the Department of Finance Canada; Chinese measures are as reported in USDA Foreign Agricultural Service summaries of State Council Tariff Commission announcements.

Phase definitions. Phase 1 (February 1–April 2) covers the two US tariff increases on China and China's product-specific tariffs of 10–15 percent, on coal, liquefied natural gas, crude oil and some machinery and vehicles from February 10 and on agricultural products from March 10. In Phase 2 (April 3–May 12) the 10 percent baseline took effect on April 5, with Canada and Mexico excluded, and higher country-specific rates took effect on April 9 and were suspended for every partner except China on April 10. Phase 3 (August 7–December 31) begins with the August framework rates. Between the May 12 and August 6 daily observations, which fall outside the phases, the dollar fell by a further 4.9 percent against the euro and 2.5 percent against the broad index; in December it was 13.1 percent weaker against the euro than in January.

Classification caveats. The US tariffs on Canada (10 percent on energy) and Mexico took effect on March 4 and from March 7 no longer applied to goods qualifying under the USMCA. Canada retaliated on March 4 with a 25 percent surtax on C$30 billion of US goods (Department of Finance Canada, 2025) and extended it to a further C$29.8 billion on March 13; Mexico did not retaliate. Canada and Mexico together carry 27.6 percent of the broad index. In the week of March 3–7 the broad index fell by 1.8 percent, more than

its net change over Phase 1, and the dollar fell by 4.4 percent against the euro. The same week also contained the March 4 announcement by Germany's prospective governing parties of a large defence and infrastructure package and a loosening of the debt brake (Ripper, 2025). The dollar strengthened against the dong by 3.2 percent between the January and December averages, all of it by July. Vietnam, however, moved during the year from a bystander-like position to a direct target: a 46 percent rate was announced on April 2 and was in force for one day before the suspension of April 10, and a 20 percent rate applied from August 7. From August 7 to November 10, Vietnam's 20 percent against China's 30 percent was relative treatment in Vietnam's favour; after November 10, when both faced 20 percent, it disappeared. Within Phase 3 the dollar was roughly flat against the dong (0.0 percent on monthly averages, +0.5 percent between end-of-month rates).

D Further Robustness

This appendix states sufficient conditions for the two-country sign result of Remark 2.1 and relaxes two maintained assumptions of the nested-CES model, linear technology and balanced trade. Throughout, e_{Aj} is units of A's currency per unit of j's currency, so a rise is a depreciation of A against j.

D.1 Two-country sufficient conditions

Let $e \equiv e_{AB}$ and let τ be A's tariff on imports from B. Border prices are one in the producer's currency, so with import and export quantities Q_M and Q_X, A's import payments are $M_A = eQ_M$, its export receipts are $X_A = Q_X$, and

$$\mathrm{TB}_A(e,\tau) \equiv X_A(e,\tau) - M_A(e,\tau),$$

where X_A and M_A include all induced effects, including income effects of tariff revenue. If TB_A is continuously differentiable with $\mathrm{TB}_{A,e} \neq 0$ at an equilibrium, the implicit function theorem gives a locally unique $e(\tau)$ with

$$\frac{de}{d\tau} = -\frac{\mathrm{TB}_{A,\tau}}{\mathrm{TB}_{A,e}}.$$

The first condition is $\mathrm{TB}_{A,\tau} > 0$: at fixed e, the tariff-induced fall in import payments exceeds any tariff-induced fall in export receipts. Because $M_A = eQ_M$ is measured at border prices, import payments fall if and only if the import quantity falls; the condition does not require import expenditure $(1+\tau)eQ_M$ to fall. Under a lump-sum rebate of revenue, the income effect can offset substitution. With ε^H the compensated own-price elasticity of import demand, η_I its income elasticity and θ the revenue share of income,

$$\left.\frac{d\log Q_M}{d\log(1+\tau)}\right|_e = -\frac{\varepsilon^H}{1-\eta_I\theta},$$

so at free trade ($\theta = 0$) the import quantity falls if and only if $\varepsilon^H > 0$. In the nested-CES model at free trade, $\varepsilon^H = \rho_D > 0$, the domestic diversion margin of Eq. (3.4), and $X_{A,\tau} = 0$ because B's income and demand are unchanged at a fixed exchange rate.

The second condition is $\mathrm{TB}_{A,e} > 0$. Since $eX_{A,e} = X_A\varepsilon_X$ and $eM_{A,e} = M_A(1-\varepsilon_M)$, with ε_X and ε_M the quantity elasticities of Section 2.1, at a balanced baseline $e\,\mathrm{TB}_{A,e} = X_A(\varepsilon_X + \varepsilon_M - 1)$, and the condition is equivalent to the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$.

When both conditions hold, $de/d\tau < 0$: the tariff appreciates A's currency against B.

These are sufficient conditions, not properties of every model: the fiscal closure (revenue wasted, spent by the government or transferred abroad) and the pricing convention (local- or dominant-currency pricing, incomplete pass-through, markups) change the quantity responses and can reverse either sign.

D.2 Sector-specific factors

Replace the linear tradable technology with a Ricardo–Viner technology, $Y_{Ti} = A_{Ti} L_{Ti}^{\gamma}$ with $0 < \gamma \leq 1$, where a fixed sector-specific factor owned by domestic households earns rents $(1-\gamma)Y_{Ti}$; non-tradables remain linear in labor, which is mobile across sectors. At a free-trade baseline, tradable employment is $L_{Ti} = \gamma\alpha_T L_i/(1-\alpha_T+\gamma\alpha_T)$ whatever the exchange rates, so the adjustment matrix is that of the linear model with the same nominal incomes. Only the disturbance changes, through the tariff-setter's income: the rebate draws labor out of tradables and raises the wage, so with s_{AB} the share of A's income spent on B's good,

$$\left.\frac{d\log I_A}{d\tau_{AB}}\right|_{\nu} = \kappa(\gamma)\, s_{AB}, \qquad \kappa(\gamma) \equiv 1 + \frac{(1-\alpha_T)^2(1-\gamma)}{\alpha_T\,(1-\alpha_T+\gamma\alpha_T)} \geq 1,$$

with equality if and only if $\gamma = 1$. Proposition 3.2 then holds with ρ_D replaced by

$$\rho_D(\gamma) \equiv \rho_D - (\kappa(\gamma)-1)m = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T)\,\frac{\gamma}{1-\alpha_T+\gamma\alpha_T} \leq \rho_D,$$

so the bystander threshold falls to $N\rho_D(\gamma)$, while the NEER still appreciates because $\rho_D(\gamma) > 0$. At the US-like parameters and $N = 3$, the threshold is 3.96 with linear technology and 3.83 at $\gamma = 1/2$.

D.3 Permanent trade imbalances

Let T_i be permanent transfers fixed in A's currency, positive for a receipt, with $\sum_{i\in\mathcal{I}} T_i = 0$. Each country's external-balance condition becomes

$$\mathrm{TB}_i + T_i = 0 \qquad \text{for every } i \in \mathcal{I},$$

so a recipient runs a trade deficit equal to its receipt. The general linear system Eq. (2.3) remains valid at any such baseline.

As an illustration of U.S. scale, let A receive a transfer equal to 3 percent of its income, paid by the target B, with the US-like parameters and $N = 3$. Because A's import share is small, financing the deficit requires a baseline appreciation of A of about 15 log points

against B. The nominal bystander threshold rises from 3.96 to 4.94: A's imports tilt toward the cheapened target, so the bystander is less exposed to diversion from A's tariff. The single-threshold structure survives on $\rho \in [1.05, 60]$. Alternative transfer allocations and tariff levels are reported in the replication package.